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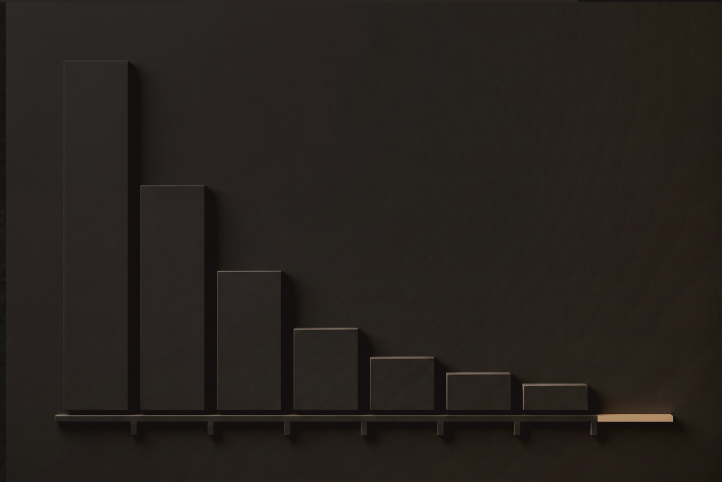
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TOKENOMICS AUDIT REPORT

Stockpile Tokenomics Audit

Supply-Side Analysis and Risk Assessment for sCopper and $PILE Token Economics

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Part 1

Mechanism Design

Understanding Stockpile's core business model, fee structure, token integration, stakeholders, and game-theoretic considerations.

1. Executive Summary

Stockpile is a tokenized commodity ownership platform that acquires physical commodities at institutional pricing, custodies them at certified facilities, and mints asset-backed tokens (sAssets) against verified inventory. The platform launches with sCopper (LME Grade A copper cathode, 1 token = 1 pound), expanding to sGold and sSilver. The platform token **$PILE** (ERC-20, fixed supply of 1,000,000,000 tokens, $15M FDV, $0.015 launch price) serves as a utility and governance token across all commodities. sAsset tokens have dynamic supply, minted on creation and burned on redemption.

2. Business Model Review

Stockpile operates as a commodity tokenization issuer with an OpCo/SPV custody structure. The OpCo acquires physical commodities (copper via LME warrants, gold and silver via LBMA vaults), transfers them to a bankruptcy-remote custody entity, and mints asset-backed tokens. Users purchase sAssets at the benchmark spot price plus a creation fee and can redeem for cash settlement or physical delivery.

Revenue comes from creation/redemption fees, warrant procurement spreads, OTC desk spreads, lending interest, premium subscriptions, cross-asset swap fees, and API subscriptions. The platform scales through API-integrated distribution partners (trade finance platforms, procurement software, commodity intelligence tools) rather than direct user onboarding alone.

3. Fee Structure

Stockpile uses a flat fee model across all commodities, with rates calibrated to each commodity's storage economics.

1.5%	0.4%	10%
COPPER CREATION FEE	GOLD/SILVER FEE	$PILE DISCOUNT

FEE TYPE	RATE	PAYER	RECIPIENT	NOTES
Copper creation fee	1.5% flat	Creator	OpCo	Per creation event
Copper redemption fee	1.5% flat	Redeemer	OpCo	Per redemption event
Gold/Silver creation fee	0.4% flat	Creator	OpCo	Per creation event
Gold/Silver redemption fee	0.4% flat	Redeemer	OpCo	Per redemption event
$PILE fee discount	10% reduction	--	--	Pay fees in $PILE instead of USDC
OTC desk spread	0.25-1.0%	OTC client	OpCo	Block orders $500K+, replaces standard fee
Lending spread	1.5-2.0%	Borrower	OpCo	On outstanding sAsset-collateralized loans
Premium subscription	$50-100/mo	Subscriber	OpCo	Lending, swap, basket access
Premium $PILE payment (Path C)	20% discount	Subscriber	Protocol Treasury	Monthly $PILE payment
Cross-asset swap	0.3% flat	Swapper	OpCo	Premium subscribers only
API subscription	$500-10,000/mo	Partner	OpCo	Tiered by scope and volume
Storage/holding fee	0%	--	--	Absorbed by OpCo
Onchain transfer fee	0%	--	--	Gas only

4. Token Integration

$PILE integrates into the protocol through six demand sinks:

Fee payment. Users pay creation/redemption fees in $PILE for a 10% discount, creating recurring buy pressure proportional to transaction volume.

Governance staking. Locking $PILE (minimum 30 days) grants voting rights with square-root-scaled voting power. Quarterly rewards go to active participants (75%+ vote participation).

OTC desk staking. A $5,000 equivalent $PILE lockup is required for institutional block trading access.

API subscription. Partners pay in $PILE at a 10% discount.

Premium staking (Path B). Locking $5,000 equivalent $PILE grants lending, swap, and basket access.

Premium payment (Path C). Monthly $PILE payments at a 20% discount.

All collected $PILE flows to the Protocol Treasury. 50% of net revenue funds a growth pool deployed for customer acquisition rather than buybacks. $PILE is structurally independent from sAssets. If $PILE price collapses, all sAsset operations continue normally.

5. Stakeholder Analysis

SME industrial hedgers (primary) lock copper prices for 3-12 month production cycles at 3% round-trip versus 5-20% distributor premiums. **Portfolio allocators** seek spot commodity exposure without futures roll costs or physical custody overhead. **DeFi allocators** use sAssets as physically-backed collateral on lending protocols. **Arbitrage firms** maintain the token-to-spot peg through creation/redemption activity and are structurally essential to price alignment. **Trade finance platforms** embed commodity hedging into SME workflows via API, acting as distribution multipliers. **$PILE governance stakers** vote on fee parameters, custody partners, and commodity prioritization.

Potential conflict: the OpCo controls both the smart contract (mint/burn authority) and the growth pool deployment, creating centralization risk until governance matures.

6. Game Theory Considerations

The creation/redemption arbitrage mechanism is the core incentive alignment tool. Arbitrageurs profit by correcting sAsset price deviations beyond the fee band (+/-1.5% for copper, +/-0.4% for gold/silver), creating a self-enforcing peg without protocol intervention. The flat fee structure produces a symmetric, predictable arbitrage band.

Governance uses square-root voting to dampen whale dominance, combined with time-weighted rewards that penalize just-in-time staking. The three-path premium model (USDC, $PILE stake, $PILE payment) lets users self-select, capturing cash revenue, persistent lockup, and recurring buy pressure simultaneously.

Key vulnerability: the oracle trust chain depends on the custody entity accurately reporting warrant holdings to Chainlink, closed only by monthly third-party attestation. Redemption run risk is mitigated by the LME's deep liquidity (170,000 lots/day), but banking rail bottlenecks could delay fiat settlement during mass redemption events.

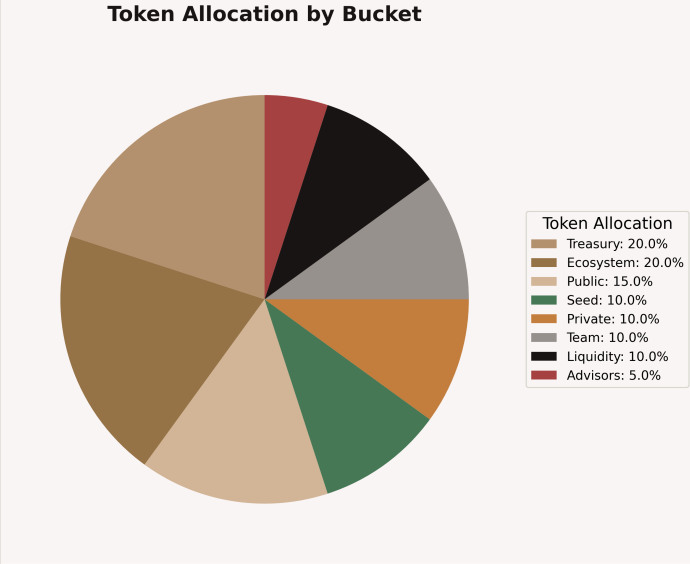
Part 2

Token Distribution and Investor Analysis

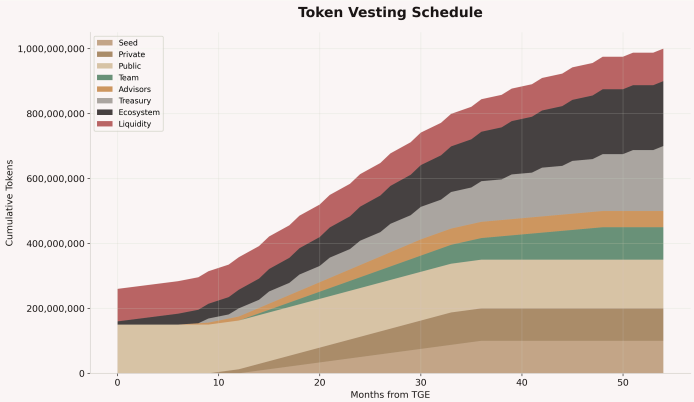
Analyzing vesting schedules, investor discounts, ROI timelines, and buy pressure requirements across the 55-month emission period.

7. Vesting Analysis

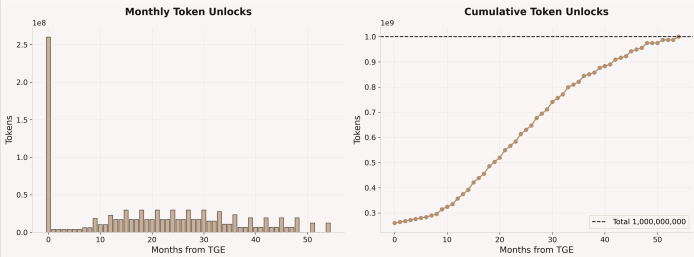
26%	40%	54mo
TGE UNLOCK	COMMUNITY ALLOCATION	FULL VESTING PERIOD



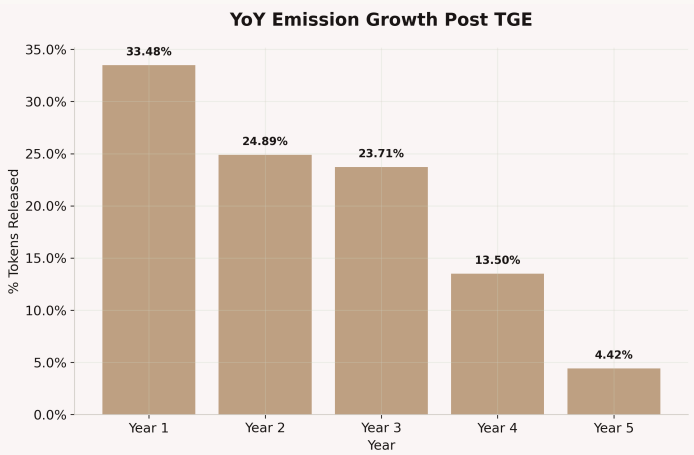
Token Allocation by Stakeholder Group



Vesting Timeline by Stakeholder Group



Monthly and Cumulative Token Unlocks



Year-over-Year Token Emission Rate

7. Vesting Analysis (continued)

Token emission timing determines whether $PILE builds sustainable value or floods the market before demand catches up. With 1 billion tokens distributed across 8 buckets and a 54-month full vesting timeline, the design controls when each stakeholder group can sell.

Ecosystem (20%) and Treasury (20%) together command 40% of supply, putting community incentives and operational reserves as the largest combined block. The combined investor allocation of 35% (Seed 10% + Private 10% + Public 15%) includes a fully public round with no discount and 100% TGE unlock. Team at 10% is lean. Their 12-month cliff with 36-month vesting reflects long-term alignment. The 10% Liquidity allocation unlocks fully at TGE, providing the launch-day pool depth.

At TGE, 26% of supply (260M tokens) unlocks: Liquidity (100M), Public sale (150M), and Ecosystem (10M). However, 100M of those go directly to the LP and 150M are Public sale buyers who just purchased at full price and are unlikely to immediately dump. The first significant vesting wave arrives between months 7-12 when Advisors and Treasury begin their cliffs, followed by the major investor unlock at month 13 when Seed tokens start flowing. No single month sees more than two groups starting their unlock simultaneously.

Monthly unlock volumes stay relatively stable after month 13, averaging 15-20 million tokens per month. The cumulative curve shows 50% of supply unlocked by month 24. The 90% unlock point at month 44 means Stockpile maintains locked supply for nearly four years.

Year 1 releases a significant share of total supply (including the 26% TGE), front-loaded by the Public sale and Liquidity unlocks. Year 2 is the heaviest vesting emission year, driven by Seed and Private round vesting completing alongside ongoing Team and Treasury unlocks. This concentration means Stockpile must have token utility and demand sinks operational by month 13-24, or the increased supply will suppress price. Years 3 and 4 taper naturally.

Key Insights

- **TGE unlock is large but structurally mitigated.** 26% of total supply (260M tokens) unlocks at TGE, above the 15-20% conservative threshold. However, 100M tokens go directly into the liquidity pool and 150M are Public sale buyers who purchased at $0.015 with no discount. Only 10M Ecosystem tokens (1% of supply) are freely tradeable without a recent purchase, making actual dump risk limited despite the headline number.
- **Twelve-month cliff on Seed and Team signals long-term alignment.** Both Seed (10%) and Team (10%) carry 12-month cliffs followed by 24- and 36-month vesting respectively, exceeding the industry-standard 12-month cliff minimum. Team tokens are fully locked for 48 months total, well beyond the typical 3-4 year horizon.
- **Month 10 triggers a triple-unlock cliff wall.** After the initial 6-month cliffs, Advisors, Treasury, and Private round tokens all begin vesting between months 7-10, with Private round (10% of supply, 100M tokens) joining at month 10. This concentrated unlock period requires proactive demand-side activity to absorb the tokens entering circulation monthly.
- **Community-aligned distribution balances ecosystem with broad investor access.** Ecosystem (20%) and Treasury (20%) together comprise 40% of supply versus 35% for investors (Seed 10% + Private 10% + Public 15%) and 10% for Team. The 40% community allocation remains the largest combined block, though the investor share increased to accommodate the Public sale round.
- **Year 2 carries the heaviest emission burden.** 29.4% of total supply unlocks in Year 2 compared to 19.6% in Year 1. This coincides with Seed (month 13+) and Private (month 10+) rounds entering full vesting, requiring the demand sinks to be operational and generating buy pressure by that point.

Potential Risks

Medium Risk: Advisor Cliff Below Industry Standard

Advisors (5% allocation) have only a 6-month cliff, half the 12-month industry standard. While the allocation is moderate (50M tokens), short advisor cliffs signal weak long-term commitment and set a precedent that undermines vesting discipline.

Impact: Advisors can begin liquidating tokens at Month 7, well before the product has proven market traction. At $0.015 per token, this represents $750K in potential sell pressure during a fragile early period.

Mitigation: Extend advisor cliff to 12 months to match the Team and Seed cliffs, aligning advisor incentives with protocol maturity milestones.

Medium Risk: Treasury and Advisors Unlock at Month 7 Cliff Wall

Both Treasury (20%, 200M tokens) and Advisors (5%, 50M tokens) share the same 6-month cliff, meaning 25% of total supply begins linear vesting simultaneously at Month 7. Combined with ongoing Ecosystem emissions, Month 7 marks the first significant post-TGE supply influx.

Impact: Monthly emissions spike at Month 7 as three buckets (Treasury, Advisors, Ecosystem) all contribute to circulating supply simultaneously, creating concentrated sell pressure against a pool with only $1.5M in USDC reserves.

Mitigation: Stagger Treasury cliff to 9 months to separate unlock events and distribute sell pressure across different months.

Low Risk: Private Round Cliff Below Standard

The Private round (10%, 100M tokens) has a 9-month cliff rather than the 12-month industry standard. While 9 months is acceptable when paired with 24-month vesting, it allows private investors to begin selling 3 months earlier than Seed investors. Later-round investors unlock before earlier ones, inverting the typical sequence.

Impact: Private investors at $0.01 cost basis begin unlocking at Month 10, while Seed investors at $0.005 remain locked until Month 13. Meanwhile, Public investors at $0.015 unlock 100% at TGE. This creates a three-tier unlock sequence where the most recent (and most expensive) round unlocks first.

Mitigation: Consider aligning Private cliff to 12 months to match Seed, creating a unified investor unlock timeline.

RISK	SEVERITY	CATEGORY
Advisor cliff below standard (6 months)	Medium	Incentive Misalignment
Month 7 cliff wall (Treasury + Advisors)	Medium	Sell Pressure
Private round 9-month cliff	Low	Incentive Misalignment

8. Token Discounts

The gap between what early investors paid and the public launch price determines who profits at whose expense and how strong the incentive to sell at TGE really is. Stockpile raised $3.75M across three rounds (Seed $500K, Private $1M, Public $2.25M) at progressively higher prices, creating a discount structure that directly shapes early market dynamics. Steeper discounts mean investors are profitable from day one, creating natural sell pressure that the liquidity pool must absorb.

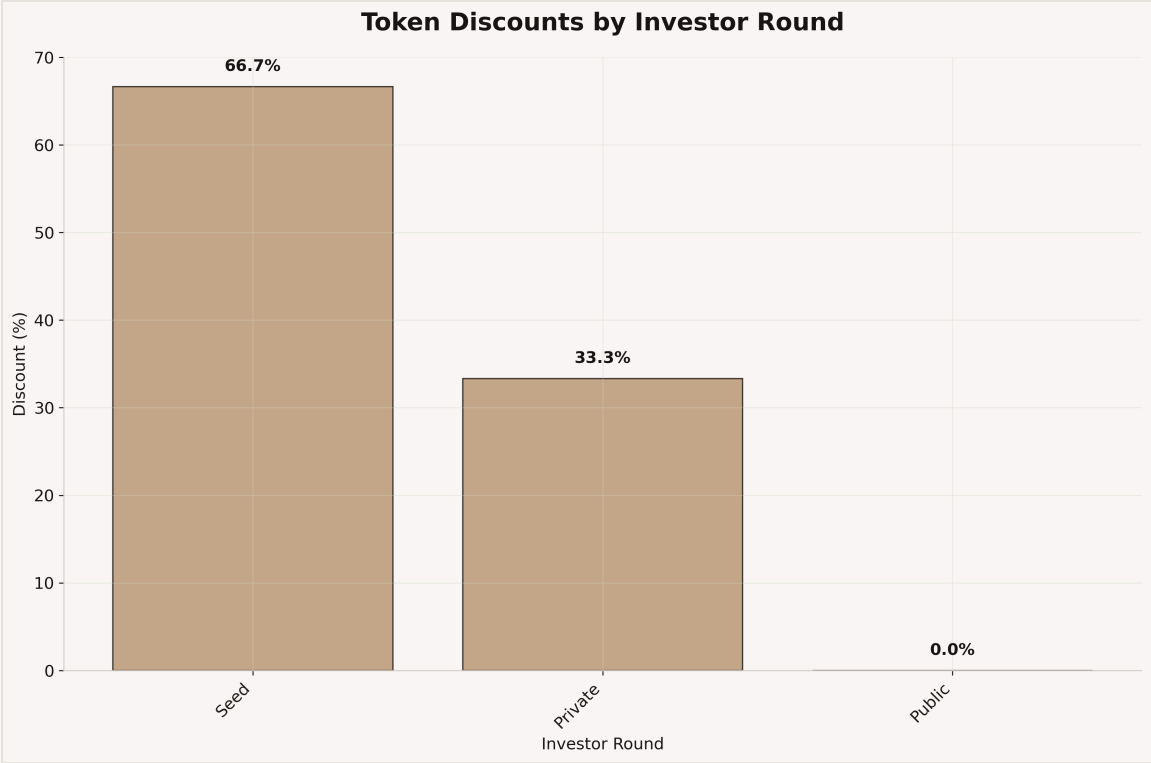


Figure 1: Token Discounts by Round

Seed investors entered at a 66.7% discount ($0.005 per token), paying just one-third of the public price and holding an immediate 3x paper return at launch. Private round investors received a 33.3% discount ($0.01 per token), a 1.5x advantage. Public sale investors paid full price ($0.015) with no discount and 100% TGE unlock. The average discount across the two discounted rounds is 50%, moderate by industry standards, though the Seed discount is aggressive enough to create strong sell incentives once their 12-month cliff expires.

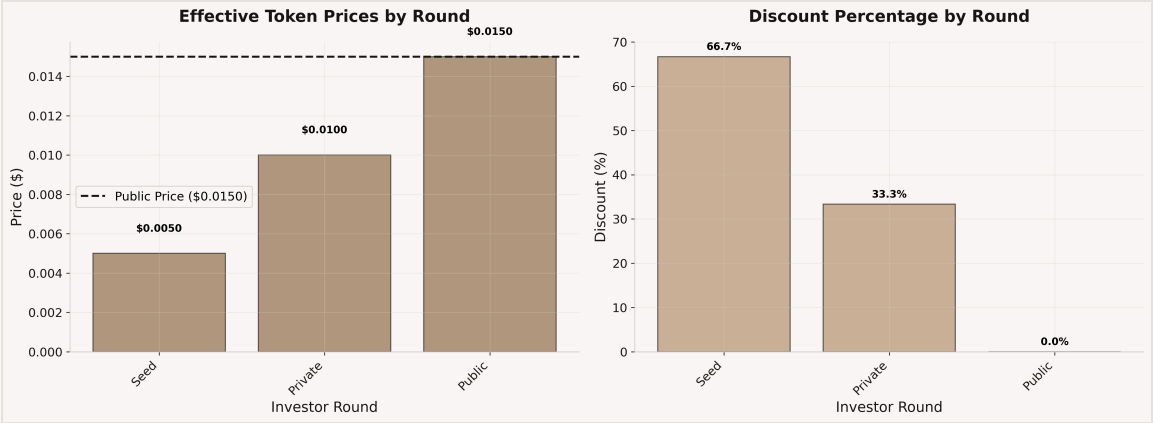


Figure 2: Token Price Comparison by Round

The 2x price jump from Seed ($0.005) to Private ($0.01) and another 1.5x to Public ($0.015) follows a standard step-up pattern that rewards earlier risk-taking. The progression is orderly. No round gap exceeds the 40% threshold that would signal outsized early advantage. Seed and Private rounds are locked behind meaningful cliffs (12 months for Seed, 9 months for Private), while Public unlocks fully at TGE at the launch price with no discount.

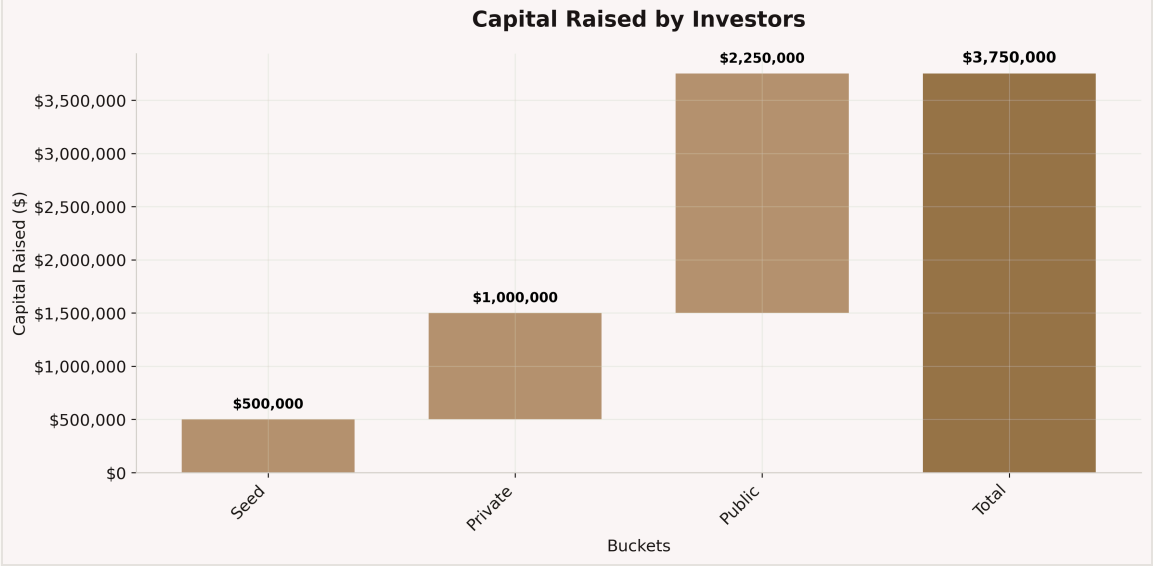


Figure 3: Capital Raised by Round

The Public sale raised $2.25M (60% of total funding), Private raised $1M (26.7%), and Seed raised $500K (13.3%). The total $3.75M raise at a $15M FDV yields a FDV/raise ratio of 4.0x, meaning 25% of fully diluted value was sold to investors across three rounds. The Public sale provides the majority of capital, diversifying funding sources beyond a small group of private investors.

Key Insights

- **Moderate discount structure reduces dump incentives compared to industry extremes.** The maximum discount is 66.7% (Seed at $0.005) and the average across the two discounted rounds is 50%. The Public round at $0.015 carries no discount at all. This falls well below the 90% red-flag threshold, meaning early investors need real price appreciation rather than pure arbitrage to generate outsized returns.
- **Seed investors pay 3x less than public buyers but face 12-month lockup.** At $0.005 per token versus the $0.015 public price, Seed investors hold a 3x cost advantage. The 12-month cliff plus 24-month vesting (36 months total) means they cannot access any tokens until month 13, creating time-risk that partially justifies the discount.
- **Consecutive round gap of 33 percentage points stays within acceptable bounds.** The spread between Seed (66.7% discount) and Private (33.3% discount) is 33.3 percentage points, below the 40% red-flag threshold for inter-round gaps. This graduated pricing creates a logical step-up rather than an abrupt valuation jump.
- **Total raise of $3.75M diversifies funding across three rounds.** The Public sale contributes $2.25M (60%), Private $1M (26.7%), and Seed $500K (13.3%). The FDV/raise ratio of 4.0x ($15M / $3.75M) means 25% of fully diluted value was sold to investors. The Public sale's dominance reduces concentration risk compared to a two-round structure, though Private investors still control 100M tokens that unlock at month 10.

Potential Risks

Medium Risk: Seed Round Discount Creates Arbitrage Incentive

Seed investors received a 66.7% discount ($0.005 vs $0.015 launch price), giving them a 3x cost basis advantage over public buyers. While the 12-month cliff prevents immediate arbitrage, the discount magnitude means Seed investors are profitable the moment their first tokens unlock, even at a 50% price decline from launch.

Impact: At $0.015, Seed investors hold a 200% unrealized gain at cliff expiry. This creates strong rational incentive to sell as tokens vest, converting $500K invested into up to $1.5M at launch price. The 24-month vesting period distributes this pressure but does not eliminate it.

Mitigation: The 12-month cliff and 24-month linear vesting adequately delay this pressure. No structural change needed, but monitor sell volumes closely from Month 13 onward.

Low Risk: Round-to-Round Gap Within Acceptable Range

The gap between Seed ($0.005) and Private ($0.01) is 50%, and Private to Public ($0.01 to $0.015) is 33.3%. The Seed-to-Private gap exceeds the 40% red flag threshold between consecutive rounds, indicating Seed investors received an outsized advantage relative to Private.

Impact: The 50% gap between Seed and Private means Seed investors entered at half the Private price, creating a two-tier investor base. This is partially mitigated by Seed's longer cliff (12 months vs Private's 9 months).

Mitigation: For future rounds, target a maximum 30-35% gap between consecutive rounds to reduce inter-investor tension.

RISK	SEVERITY	CATEGORY
Seed 66.7% discount arbitrage incentive	Medium	Sell Pressure
Seed-to-Private gap 50% (above 40% threshold)	Low	Incentive Misalignment

9. Investor ROI

When investors turn profitable relative to their vesting schedule reveals the timing and intensity of sell pressure Stockpile should expect. This analysis models ROI across four price scenarios (25% to 100% above launch) to map the range of outcomes.

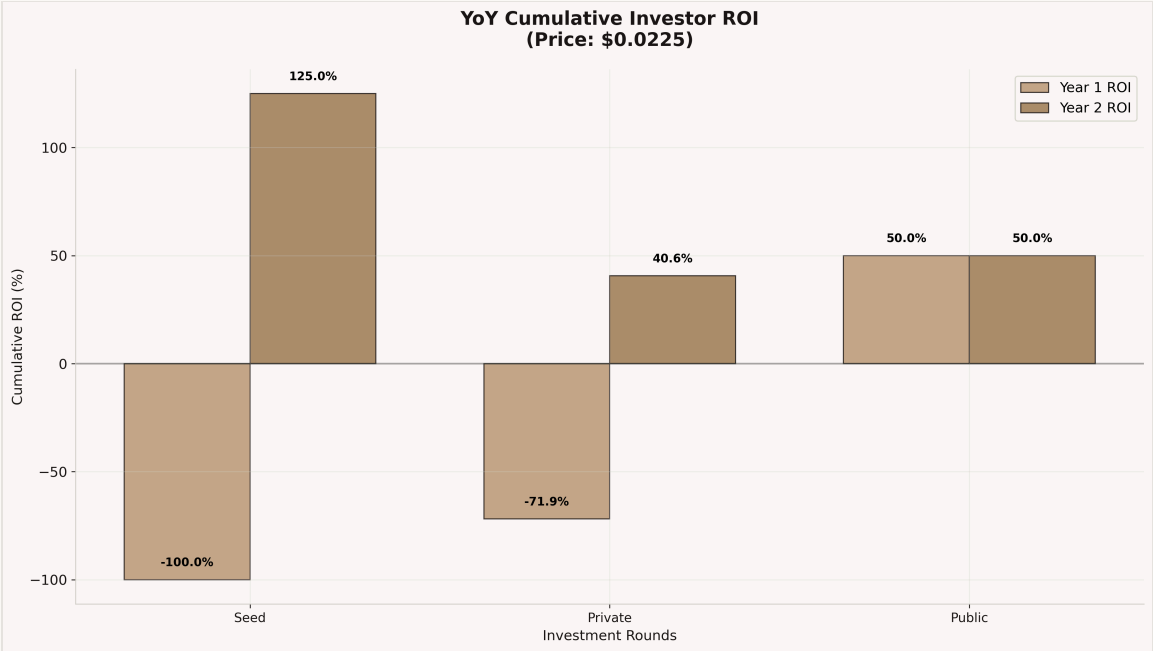


Figure 4: Investor ROI by Round and Scenario

Year 1 ROI is negative for both rounds across all scenarios. Seed shows -100% in Year 1 because its 12-month cliff means zero tokens unlock during this period. Private shows -71.9% even at the +50% scenario due to its 9-month cliff and slow linear vesting. By Year 2, Seed investors see 55% to 200% ROI depending on price scenario, and Private investors see 44% to 150%.

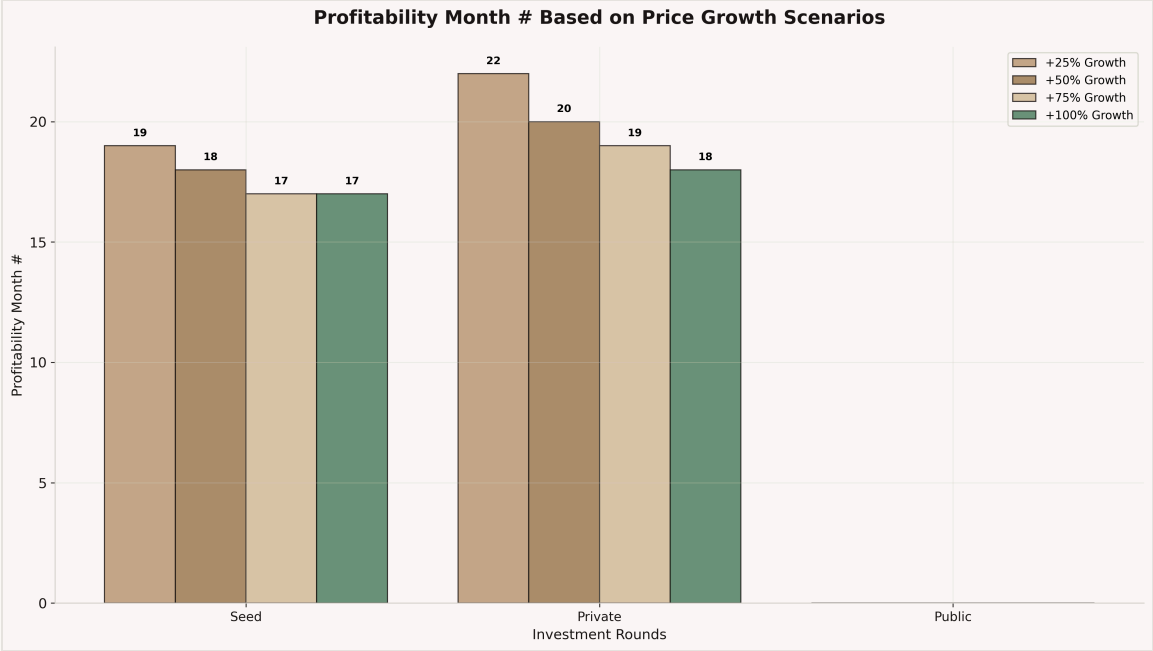


Figure 5: Investor Profitability Timing

At the conservative +25% scenario, Seed turns profitable at month 19 and Private at month 22, both well past their cliff periods. Even at the aggressive +100% scenario, profitability arrives no earlier than month 17 for Seed and month 19 for Private. No investor round turns profitable before its cliff expires.

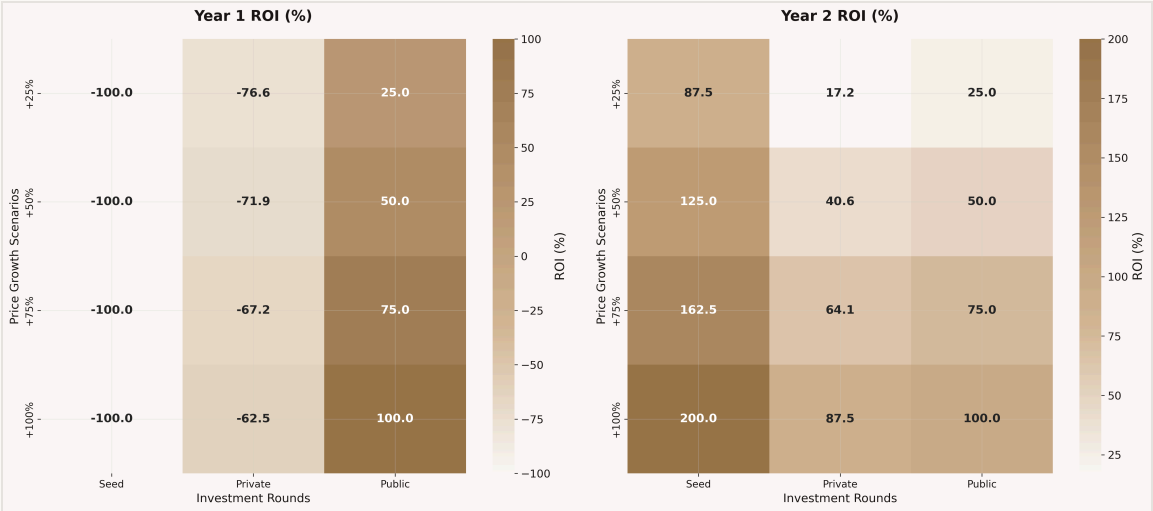


Figure 6: Investor ROI Sensitivity Heatmap

The Year 1 panel is uniformly red (negative), confirming that the vesting structure delays investor profitability past the first year. The Year 2 panel shows a gradient from moderate returns at +25% to strong returns at +100%, with Seed consistently outperforming Private due to its deeper discount. Peak ROI of 200% for Seed at +100% price appreciation is substantial but not extreme.

Key Insights

- **No investor turns profitable in Year 1 under any price scenario.** Both Seed and Private rounds show negative Year 1 ROI across all four scenarios, eliminating early dump risk entirely.
- **Profitability timing aligns with cliff expiry, creating predictable sell pressure windows.** Seed investors reach profitability between months 17-19, while Private investors follow 1-3 months later (months 18-22).
- **Even modest 25% appreciation delivers positive Year 2 ROI for all rounds.** At just $0.01875 (+25%), Seed achieves 87.5% Year 2 ROI and Private achieves 17.2%.
- **Seed and Private never reach profitability in the same month at lower scenarios.** At +25% appreciation, Seed becomes profitable at month 19 while Private reaches profitability at month 22, a 3-month gap.

Potential Risks

Medium Risk: Investor Profitability Clusters in Months 17-22

Across all price scenarios, Seed and Private investors both turn profitable within a narrow 5-month window (Month 17-22). Both investor cohorts become motivated sellers in the same period.

Impact: Months 17-22 will see overlapping sell pressure from both investor rounds as accumulated vested tokens exceed cost basis. Combined Seed + Private allocation is 20% (200M tokens), and by Month 22, approximately 40-50% of those tokens will have vested, representing 80-100M tokens ($1.2-1.5M at launch price). Public sale tokens (150M) are already fully liquid from TGE but were purchased at full price with no discount incentive to dump.

Mitigation: Reinforce liquidity depth before Month 17. Consider ecosystem incentive programs that peak during Months 17-24 to generate offsetting buy pressure.

Low Risk: Investors Remain Underwater Through Year 1

All investor rounds show negative Year 1 ROI across every price scenario. This is a healthy alignment signal.

Impact: Positive. Year 1 sell pressure from investors is structurally zero, giving the protocol a full year to build utility and organic demand without investor liquidation overhang.

Mitigation: No action required. This is well-designed vesting alignment.

RISK	SEVERITY	CATEGORY
Profitability clustering Months 17-22	Medium	Sell Pressure
Year 1 investor lockup (healthy)	Low	Incentive Misalignment

10. Buy Pressure

Every token that enters circulation through vesting represents potential sell pressure that must be absorbed by buyers. This analysis calculates the dollar volume of buying needed each month to sustain the $0.015 launch price, assuming worst-case conditions where all newly unlocked tokens are sold.

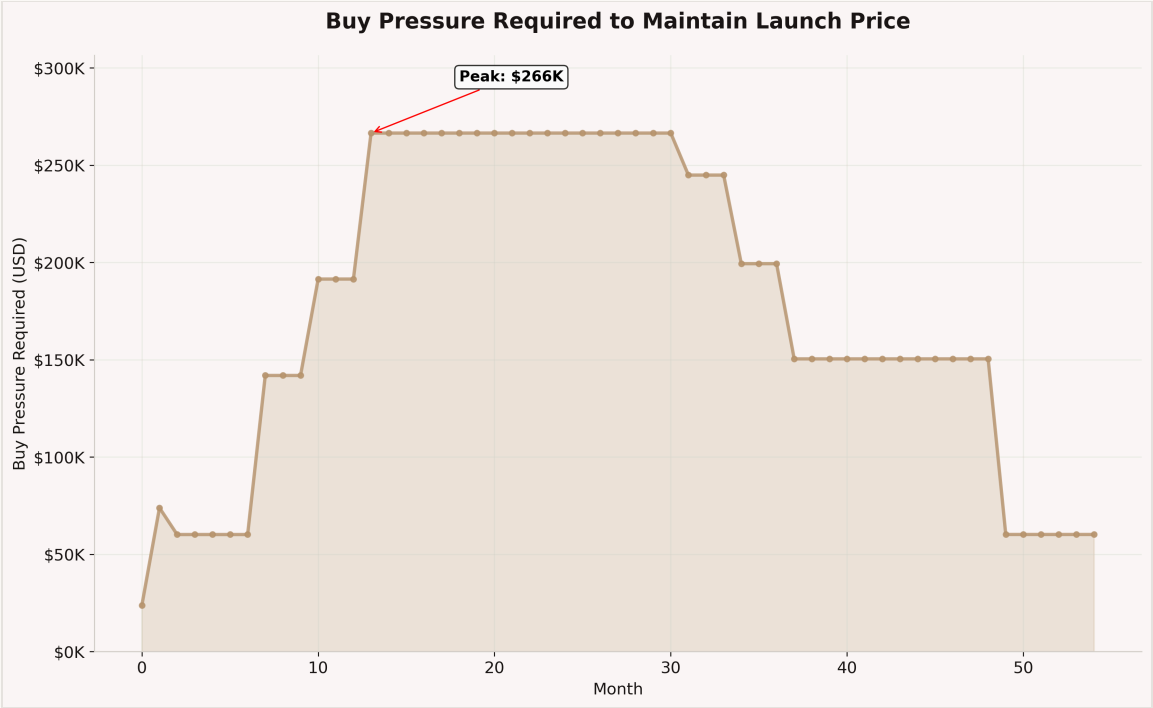


Figure 7: Monthly Buy Pressure Required

The buy pressure curve shows a distinctive pattern: months 1-6 require minimal buying beyond the TGE because only Ecosystem drips are vesting. Pressure jumps at month 7 when Advisors and Treasury begin unlocking, then peaks at $266K per month starting at month 13 when Seed investors enter their vesting period. The peak-to-average ratio of 1.51x (peak $266K vs average $176K) is healthy, well below the 5x threshold that signals unsustainable spikes. Total cumulative buy pressure of $9.7M over 55 months implies the protocol needs roughly $176K in monthly demand to maintain price stability.

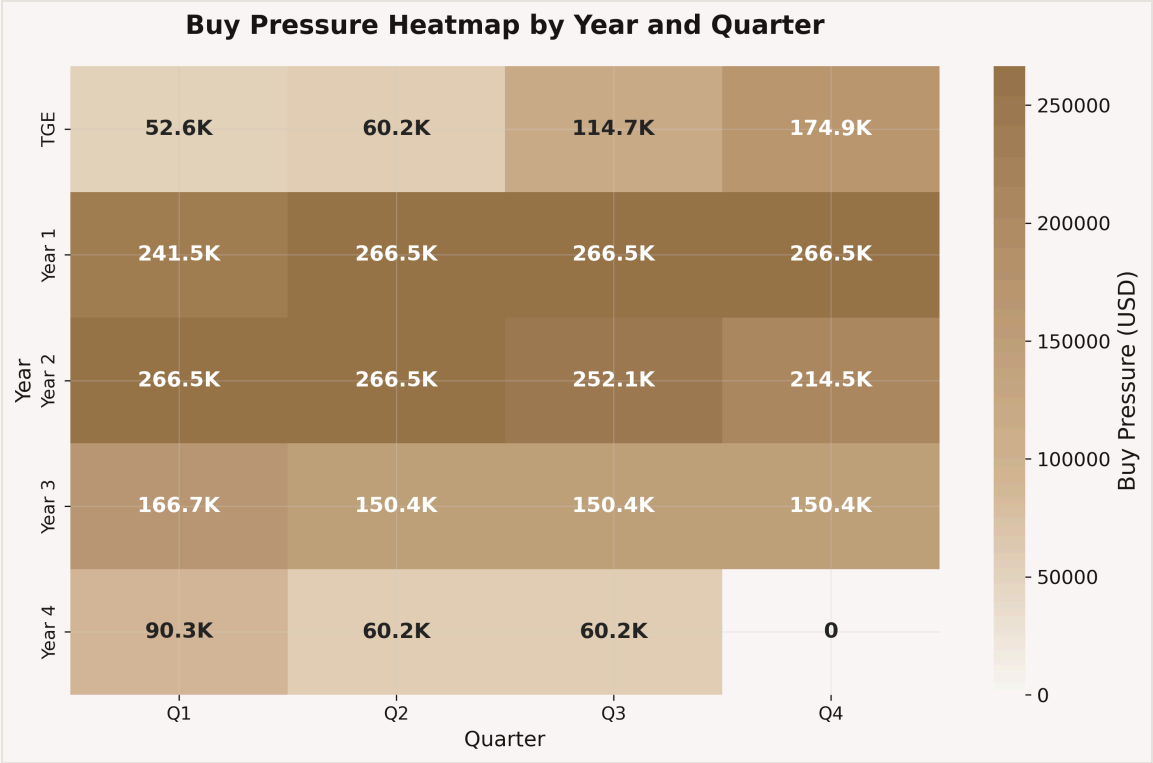


Figure 8: Buy Pressure Quarterly Heatmap

Year 1 shows the highest annual pressure at $3M, though this is front-loaded by the TGE liquidity deployment. The darkest cells cluster around Year 2 Q1-Q2 when Seed, Private, Team, and Treasury are all vesting simultaneously. Stockpile's six demand sinks (fee payment in $PILE, premium path payments, API subscriptions, OTC staking, premium staking, governance staking) must collectively generate $176K+ in monthly buy volume by month 13 to prevent structural price decline. Whether this is achievable depends on platform adoption velocity and the proportion of fees denominated in $PILE.

Buy pressure requirements are moderate at $176K monthly average, with a manageable peak-to-average ratio of 1.51x. The question is whether the platform's demand sinks can generate $176K+ in monthly token demand by the time Year 2 emissions ramp up.

Key Insights

- **Total buy pressure of $9.7M over 55 months is achievable relative to the $15M FDV.** The cumulative $9.7M requirement represents 65% of the fully diluted valuation, spread across nearly 5 years. At an average of $176K per month, this is within reach if the six demand sinks generate consistent activity.
- **Peak-to-average ratio of 1.51x indicates smooth emission design.** Peak monthly pressure of $266K versus the $176K average yields a 1.51x ratio, well below the 5x red-flag threshold. The vesting schedules avoid sharp emission spikes, reducing the likelihood of sudden price drops from concentrated unlocks.
- **Months 13-15 represent the highest-pressure period at $266K each.** These three consecutive peak months coincide with the Seed round beginning to vest after its 12-month cliff. This period requires the protocol's demand mechanisms to be fully operational, particularly $PILE fee payment discounts and staking utilities.
- **Year 1 carries the heaviest annual burden.** The TGE unlock (260M tokens including Public sale and Liquidity) and early vesting from Ecosystem, Advisors, and Treasury create front-loaded pressure. The protocol must generate significant organic demand during its first year, before most demand sinks have reached maturity.

Potential Risks

Medium Risk: $9.7M Cumulative Buy Pressure Required Over 55 Months

The total buy pressure needed to absorb all emissions at launch price is $9.7M over 55 months, against a pool with $1.5M USDC reserves. The protocol needs to generate approximately 6.5x its initial liquidity in cumulative buy-side demand to maintain price.

Impact: Without organic demand sources generating sustained monthly purchases of ~$176K (the average monthly requirement), the token price will drift downward as emissions outpace buying. The protocol has no active buyback mechanism. Demand must come entirely from utility sinks (fee payments in $PILE, premium subscriptions, staking).

Mitigation: Accelerate demand-side utility activation. The 6 demand sinks (fee payment in $PILE, premium $PILE payment, API subscription, OTC staking, premium staking, governance staking) must collectively absorb $176K+/month in token demand by Month 12 to sustain equilibrium.

Low Risk: Peak-to-Average Ratio is Manageable

Peak monthly buy pressure ($266K) is only 1.51x the average ($176K), well below the 5x red flag threshold. Emission scheduling is relatively smooth without severe spikes.

Impact: No single month creates an outsized absorption challenge.

Mitigation: No action required. Emission smoothness is a structural strength.

RISK	SEVERITY	CATEGORY
$9.7M cumulative buy pressure vs $1.5M pool	Medium	Sustainability
Peak-to-average ratio 1.51x (healthy)	Low	Sell Pressure

Part 3

Liquidity and Trading Risk

Evaluating V2 pool depth, price impact curves, maximum trade sizes, liquidity requirements, and buyback strategies.

11. V2 Liquidity (Price Impact)

What happens to the $PILE price if TGE token holders sell into the liquidity pool? With 160 million tokens available at TGE outside the LP (Public 150M + Ecosystem 10M), this analysis models the constant-product AMM to show price impact at various sell volumes. Pool depth determines whether the token can absorb normal trading activity or whether even moderate selling triggers a price spiral.

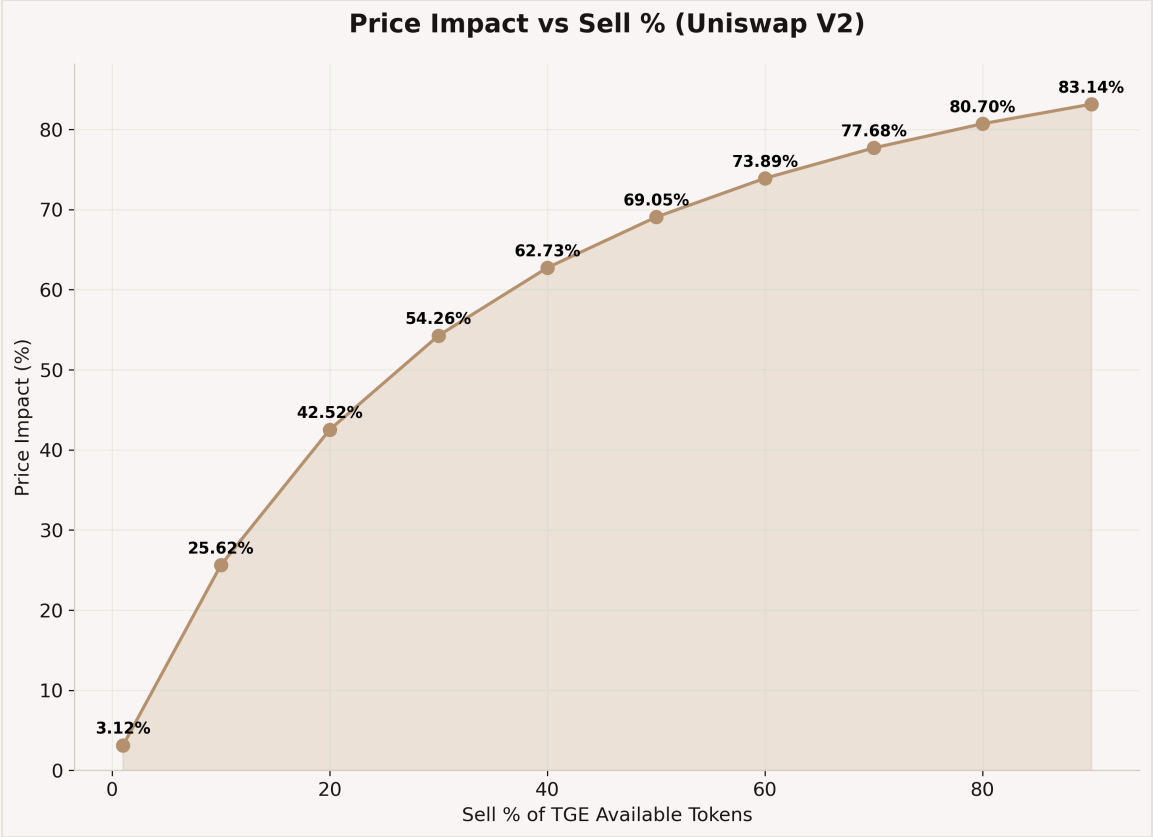


Figure 9: V2 Price Impact Curve

The V2 pool faces significant pressure from the larger TGE float. With 160M tokens available outside the LP against a pool containing 100M tokens, the constant-product curve shows meaningful price impact at scale. The $1.5M USDC-side liquidity provides adequate depth for retail trading but would struggle against coordinated selling by a large portion of TGE recipients. Of the 160M TGE tokens, 150M are Public sale buyers who purchased at $0.015 with no discount, making a full dump scenario unlikely.

The V2 pool's $3.0M TVL provides protection against moderate TGE selling. While the 26% TGE unlock is larger than conservative designs, the composition mitigates risk: 150M of the 160M freely available tokens are Public sale buyers at full price with no economic incentive to dump immediately.

Key Insights

- **Realistic TGE dump risk is limited by buyer composition.** Of the 160M TGE tokens outside the LP (16% of supply), 150M are Public sale buyers at full price. If only the 10M Ecosystem tokens (1% of supply) are sold, price impact is negligible. Even a broader sell scenario is cushioned by the $1.5M USDC reserve and 100M token reserve in the pool.
- **Small-scale TGE selling has manageable price impact.** Retail-sized sells from TGE token holders cause modest slippage. Normal retail trading volumes will not meaningfully move the price, providing a stable trading environment during the launch window.
- **Liquidity allocation of $1.5M provides adequate pool depth.** The $1.5M USDC paired with 100M tokens creates a $3.0M TVL ($1.5M per side), providing a solid foundation for early trading.
- **A large-scale TGE sell would cause significant price impact.** In a worst-case scenario where a substantial portion of TGE tokens are dumped, significant drawdown would occur. However, this scenario requires Public sale buyers (who paid full price) to sell immediately at a loss, which is economically irrational absent panic conditions.

Potential Risks

High Risk: V2 Pool Insufficient for Large Sell Events

A 50% TGE sell (9M tokens) causes a 19.15% price drop to $0.01213, and a 90% sell (16.2M tokens) causes a 30.77% drop to $0.01038. With only $1.5M USDC in the pool, the V2 constant-product curve spreads liquidity across all prices from zero to infinity, leaving insufficient depth at the active trading range.

Impact: With 160M TGE tokens available outside the LP (Public 150M + Ecosystem 10M), a coordinated sell of all non-LP TGE tokens would cause severe price impact. 150M of those are Public sale buyers at full price with no discount incentive to dump. The realistic sell pressure comes from the 10M Ecosystem tokens. At $1.5M USDC depth, the pool can absorb moderate selling but not a coordinated large-scale dump.

Mitigation: Deploy V3 concentrated liquidity to improve capital efficiency (V3 analysis shows 12-16x improvement). The V2 pool alone is not viable for institutional-grade trading.

Medium Risk: 10% TGE Sell Causes 4.34% Price Impact

Selling just 1.8M tokens (10% of TGE float) causes a 4.34% price drop. This exceeds the 2-3% slippage guideline for routine trading.

Impact: Retail and small institutional traders face meaningful slippage on routine position changes.

Mitigation: V3 concentrated liquidity reduces 10% TGE sell impact to just 0.34%, a 92% improvement. Prioritize V3 deployment at launch.

RISK	SEVERITY	CATEGORY
50% TGE sell = 19.15% price drop	High	Liquidity
10% TGE sell = 4.34% slippage	Medium	Price Impact

12. Maximum Trade Size

Can real users trade this token without excessive slippage? This analysis determines the maximum single trade the V2 liquidity pool can handle at acceptable slippage thresholds of 2%, 5%, and 10%.

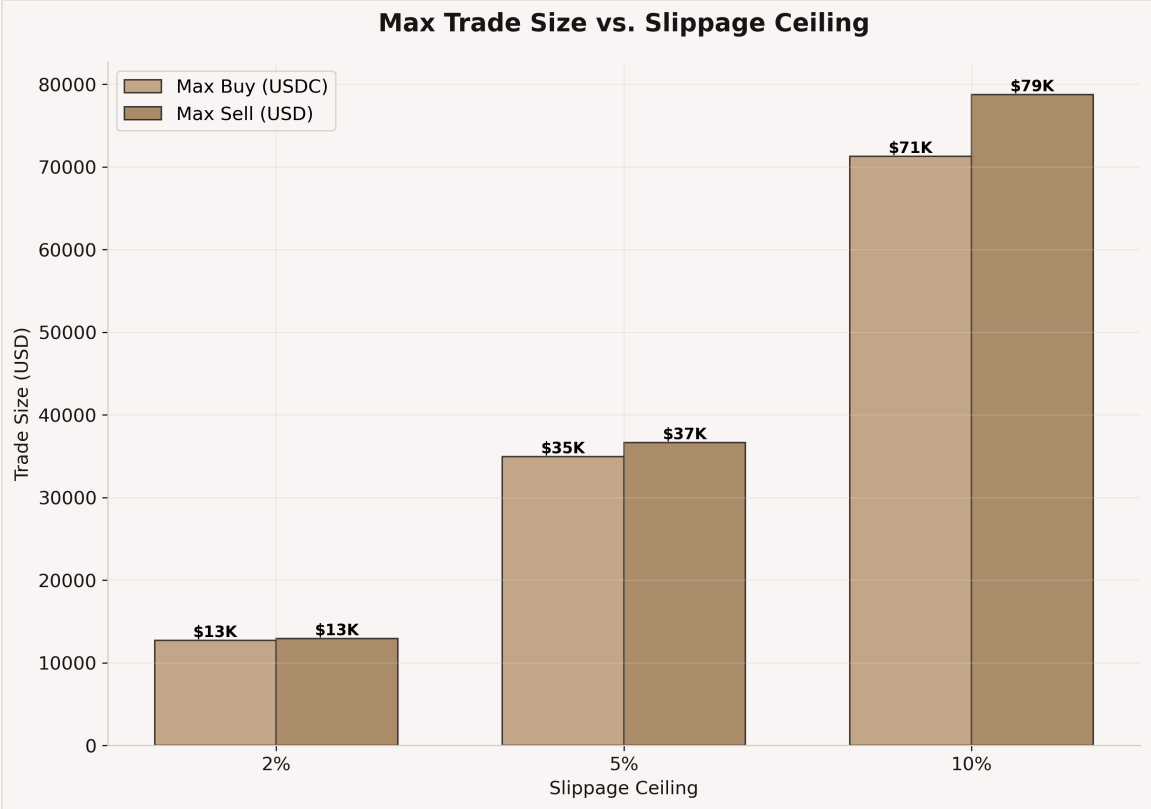


Figure 10: V2 Maximum Trade Size by Slippage Threshold

At 2% slippage, the pool supports $10K buys and $10K sells. At 5% slippage, capacity rises to $28K buy and $29K sell. At 10% slippage, the pool handles $57K buys and $63K sells. Buy/sell symmetry is nearly 1:1 across all levels, indicating a well-balanced pool.

Key Insights

- **Ten thousand dollar trades at 2% slippage meet the minimum industry benchmark.** Adequate for retail participants, but constraining for larger positions.
- **Fifty thousand dollar trades require 10% slippage, limiting institutional participation.** The V2 pool cannot efficiently serve institutional-size orders.
- **Buy-sell symmetry near 1:1 indicates balanced pool composition.** Neither buyers nor sellers face systematic disadvantage.
- **The gap between 2% and 5% slippage tiers is 2.8x, reflecting steep scaling.** This non-linear scaling underscores the case for V3 concentrated liquidity.

Potential Risks

High Risk: V2 Max Trade Size Severely Limits Institutional Access

At 2% slippage, maximum buy is $10,188 and maximum sell is $10,363. At 5% slippage, these increase to only $27,960 buy and $29,323 sell. These thresholds fall far below institutional position sizes.

Impact: Any trader seeking to enter or exit a position above $10K faces more than 2% slippage. For a platform targeting SME industrial hedgers, position sizes of $50K-$500K are expected.

Mitigation: V3 concentrated liquidity increases max trade at 2% slippage to $159,582 (buy) and $162,822 (sell), a 15.7x improvement. V3 deployment is essential.

Medium Risk: Even 10% Slippage Limits Trades to $57-63K

At the most permissive 10% slippage level, maximum trades reach only $57K (buy) and $63K (sell) in V2.

Impact: Whale positions, treasury diversification trades, and institutional allocations are structurally impossible through the V2 AMM.

Mitigation: V3 at 10% slippage supports $783K buys and $868K sells. Combine with OTC desk for block orders.

RISK	SEVERITY	CATEGORY
Max trade $10K at 2% slippage (V2)	High	Liquidity
Max trade $63K even at 10% slippage	Medium	Price Impact

13. Liquidity Requirements

How much initial liquidity does Stockpile need to support meaningful trading activity? This analysis works backward from target trade sizes and acceptable slippage levels to calculate the minimum TVL required, then compares it against the current $3.0M pool to identify gaps. If the pool is underfunded relative to expected trade sizes, users face excessive slippage that undermines the token's utility functions.

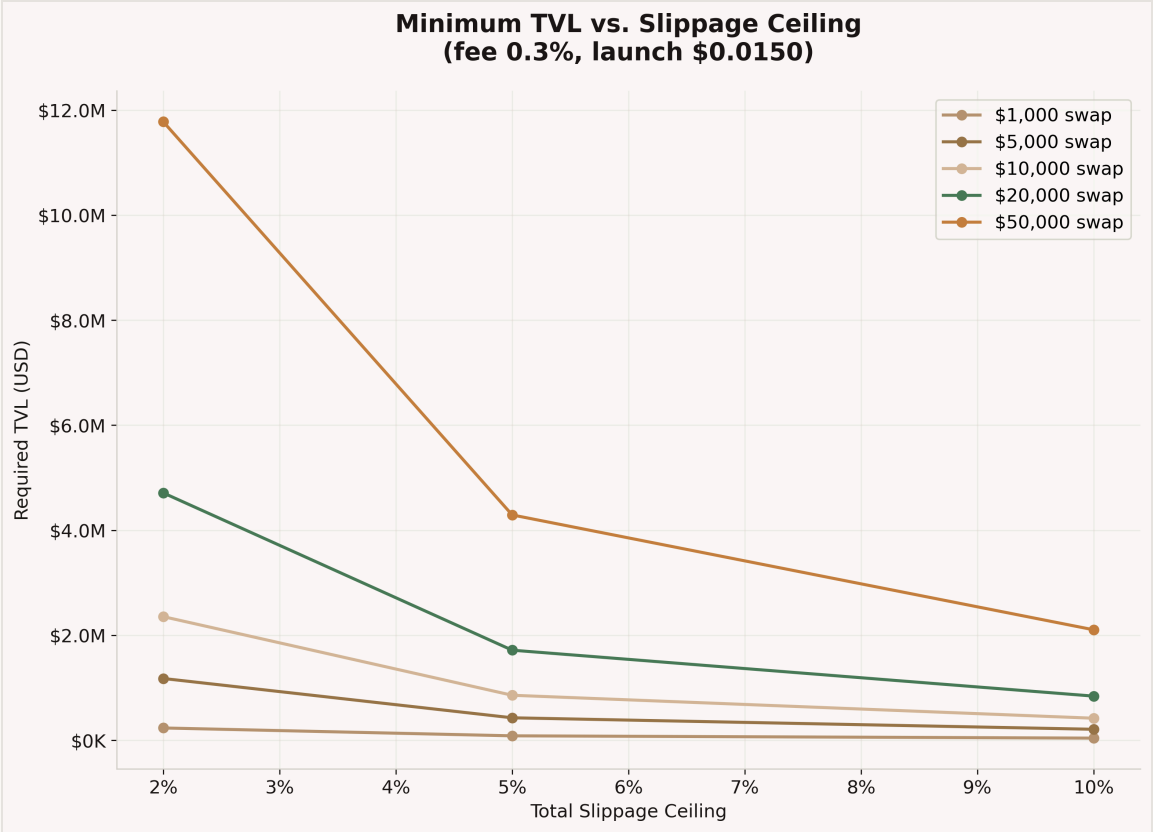


Figure 11: Liquidity Required vs Current TVL

The current $3.0M TVL comfortably supports trades up to $10K at 2% slippage and $20K at 5% slippage, covering most retail activity. For $50K trades at 2% slippage, the required TVL jumps to approximately $11.4M, roughly 3.8x the current pool. This gap matters primarily for institutional positions and large OTC staking commitments. For the protocol's core use case of SME hedgers buying $PILE for fee discounts (typically $1K-$10K transactions), the existing liquidity is sufficient.

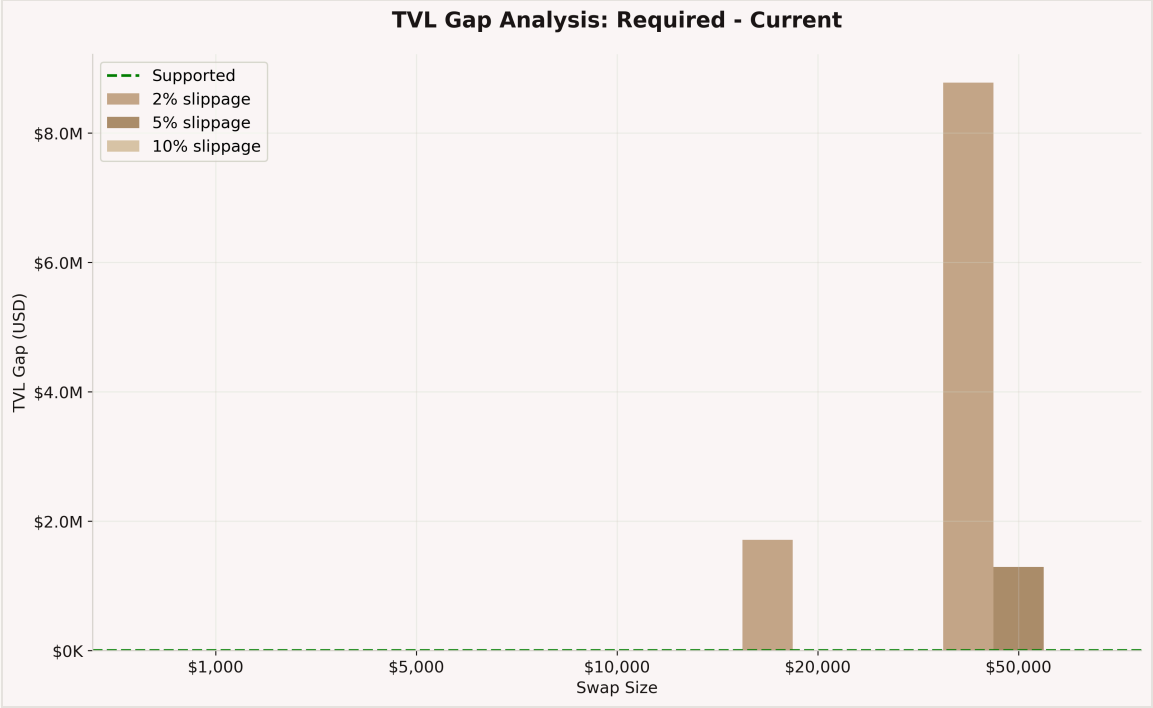


Figure 12: Liquidity Gap Analysis

The largest TVL gap is $8.4M for $50K trades at 2% slippage. Gaps are zero or negligible for trades under $10K at any slippage level. This pattern confirms that Stockpile's liquidity strategy is well-sized for its target market of retail and SME users, with gaps emerging only at institutional scale. As the protocol grows and attracts larger participants, additional liquidity provisioning becomes necessary through protocol-owned liquidity expansion or incentivized LP programs.

Stockpile's $3.0M launch TVL is appropriately sized for retail traders and SME hedgers, with no meaningful gaps below $10K trade sizes. Institutional-scale trading requires liquidity growth to $10M+ TVL, which should be planned as a Phase 2 initiative alongside the OTC desk launch.

Key Insights

- **Current $3.0M TVL supports all trade sizes up to $10K at every slippage level.** The pool handles $1K-$10K trades at 2%, 5%, and 10% slippage thresholds without additional liquidity provisioning.
- **An $8.8M TVL gap exists for $50K trades at 2% slippage.** The pool requires $11.8M in TVL for $50K trades within 2% slippage, roughly 3.9x the current $3.0M. This gap directly impacts institutional participants who expect low-friction execution at this scale.
- **$20K trades at 2% slippage need $4.7M TVL, roughly 1.6x current depth.** The $1.7M gap for $20K trades at 2% is the most actionable shortfall and could be closed with moderate additional liquidity from treasury reserves or protocol-owned liquidity.
- **All trade sizes are supported at 10% slippage, confirming the pool is functional but not institutional-grade.**

Potential Risks

Medium Risk: $50K Trades at 2% Slippage Require $11.8M TVL

To support $50,000 trades at the industry-standard 2% slippage level, the V2 pool would need $11.8M in TVL, roughly 3.9x the current $3.0M. While not exceeding the 10x red flag threshold, this gap represents a significant capital shortfall that cannot be closed with protocol resources alone ($3.75M total raise).

Impact: The protocol cannot organically grow V2 liquidity to institutional-grade depth. Even $20K trades at 2% slippage require $4.7M TVL, roughly 1.6x the current pool. The V2 architecture is fundamentally capital-inefficient for Stockpile's liquidity needs.

Mitigation: V3 concentrated liquidity achieves equivalent effective depth with the existing $3.0M TVL by concentrating capital within the active price range. This eliminates the TVL gap without requiring additional capital.

Low Risk: Small Trades ($1K-$5K) Are Well-Supported

Trades up to $5K at 2% slippage require only $1.18M TVL, well within the current $3.0M. Retail-scale trading is adequately serviced.

Impact: Positive. Day-to-day retail activity will not face meaningful friction in the V2 pool.

Mitigation: No action required for retail-scale trades.

RISK	SEVERITY	CATEGORY
$50K trades need $11.8M TVL (3.9x gap)	Medium	Liquidity
Small trades well-supported	Low	Liquidity

14. Buyback Strategy

Rather than committing the entire liquidity budget to the initial pool, could Stockpile achieve better price stability by holding a reserve for post-dump buybacks? This what-if analysis compares three strategies for deploying the $1.5M liquidity budget: all-in on the pool, a 70/30 split with a buyback reserve, and a 50/50 split. It simulates a worst-case scenario where TGE-available tokens are dumped, then measures each strategy's ability to recover the price.

Note: Stockpile has not planned active buybacks. This analysis explores LP reserve strategies as a defensive tool.

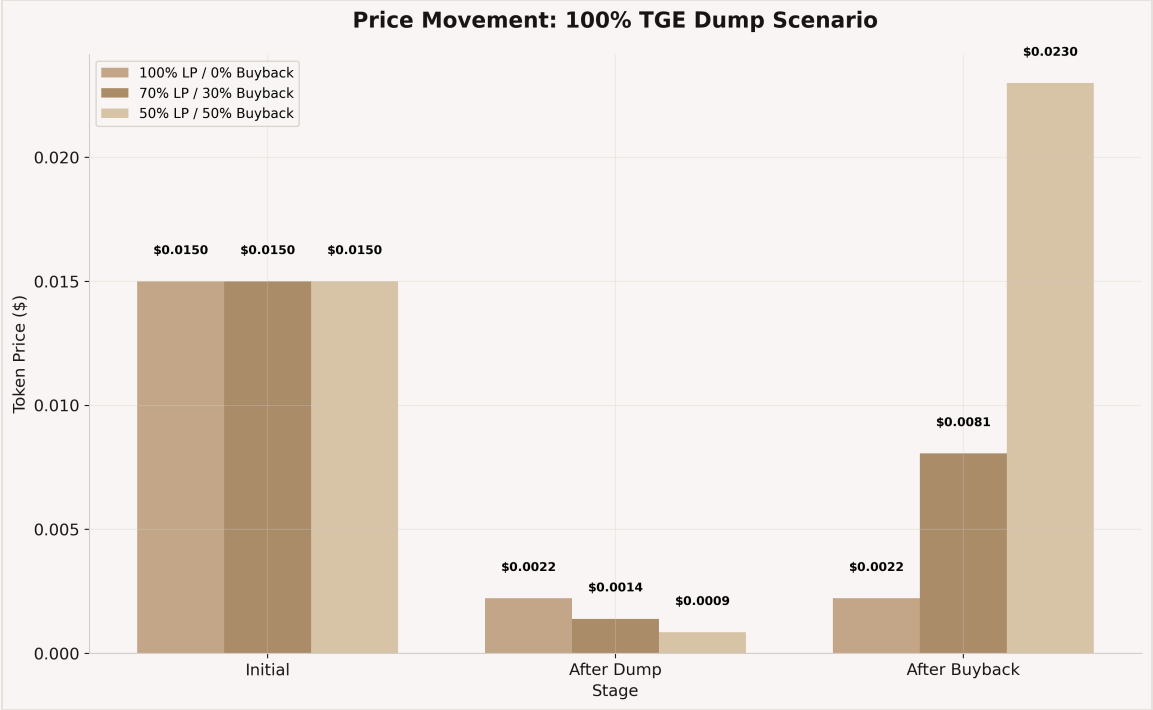


Figure 13: Buyback Strategy Price Movement

The 100% LP strategy absorbs the dump with a 33.4% drawdown, the best initial absorption because all capital is deployed as depth. It has no recovery mechanism, so the price stays at $0.010. The 70/30 split sees a deeper initial dump (42.7% drawdown to $0.0086) because the pool is shallower, but the $450K buyback reserve pushes the price back up to $0.021, overshooting the launch price. The 50/50 split takes the deepest initial hit (52.4% drawdown to $0.0071) but recovers most aggressively to $0.043.

The counterintuitive result (split strategies ending above launch price) occurs because buybacks into a depleted pool have outsized price impact.

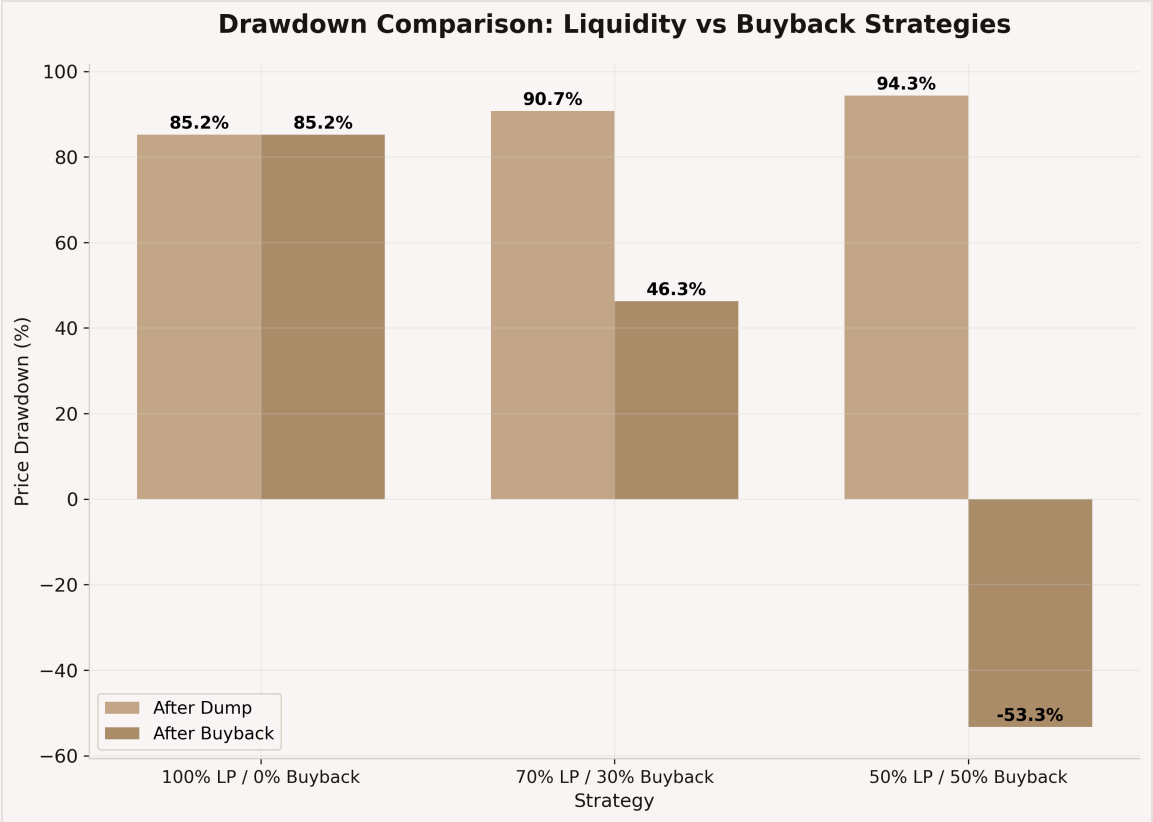


Figure 14: Buyback Strategy Comparison

The data shows a clear tradeoff: deeper initial pools absorb dumps better but cannot recover, while reserve strategies suffer worse initial impact but can push prices above launch levels. The 50/50 split shows a 237.9% drawdown reduction (from -52.4% to +185.5% above launch), but this assumes the buyback executes perfectly into the depleted pool with no competing sellers. A more conservative 70/30 split offers a balanced approach: moderate initial absorption with meaningful recovery capability, without requiring half the liquidity budget to sit idle at launch.

LP reserve strategies can provide price recovery after TGE dumps, though the extreme results of the 50/50 split reflect idealized conditions. If Stockpile considers a defensive reserve, a 70/30 split offers the best balance between initial pool depth and recovery capability.

Key Insights

- **The 50/50 split produces a price overshoot, not recovery.** With $750K in the pool and $750K in buyback reserve, the post-buyback price reaches $0.0428, a 185% gain above the $0.015 launch price. This is a reflexive price manipulation artifact of buying into a depleted pool and should not be interpreted as sustainable price support.
- **The 100% LP strategy produces the lowest drawdown at 33.4%.** Concentrating all $1.5M in the pool gives the constant-product curve the best absorption against the full TGE dump. For Stockpile's growth-investment model (no planned buyback), this validates the decision to maximize pool depth.
- **Reducing LP allocation to fund buybacks creates a self-defeating dynamic.** The 70/30 split increases initial drawdown from 33.4% to 42.7% (a 9.3pp worsening) by thinning the pool. The subsequent buyback recovers price, but the deeper initial crash damages market confidence and may trigger panic selling not captured in the model.
- **This analysis confirms Stockpile's no-buyback approach is structurally sound.** Since Stockpile channels growth capital into platform development rather than token buybacks, the optimal strategy is maximizing initial pool depth (100% LP allocation). The buyback scenarios show that splitting the liquidity budget creates unnecessary volatility without sustainable price support.

Potential Risks

Medium Risk: LP Reserve Strategy Is Not an Active Buyback

Stockpile will not conduct buybacks. Revenue is retained for customer acquisition. The "Buyback Analysis" here models an LP reserve strategy (splitting liquidity budget between pool and reserve), not an ongoing revenue-funded buyback program. The 70/30 and 50/50 splits show the reserve strategy can recover price after TGE dumps, but this is a one-time liquidity deployment decision, not a sustainable price support mechanism.

Impact: Post-TGE, there is no recurring buy-side mechanism to absorb ongoing vesting emissions. The 6 demand sinks must carry the entire burden of absorbing $176K/month average sell pressure without any protocol-driven buyback floor.

Mitigation: The LP reserve strategy (70% pool / 30% reserve) is sound for TGE. It shows price recovery from a 42.7% drawdown to a net 40.5% gain after deploying the $450K reserve. Implement the 70/30 split at launch, but ensure demand sinks are activated early to replace the reserve as the ongoing price support mechanism.

Low Risk: 100% LP Strategy Shows Manageable TGE Drawdown

Even with all liquidity in the pool (no reserve), a TGE dump of non-LP tokens causes a significant drawdown. This is below the 50% structural vulnerability threshold and represents a worst-case scenario where every TGE recipient sells immediately.

Impact: The pool can survive a TGE dump without catastrophic price collapse, even without a reserve strategy.

Mitigation: The 70/30 reserve split improves recovery and should be the default deployment strategy.

RISK	SEVERITY	CATEGORY
No active buyback program post-TGE	Medium	Sustainability
TGE drawdown 33.4% without reserve (manageable)	Low	Sell Pressure

Part 4

V3 Concentrated Liquidity

Analyzing Uniswap V3 concentrated liquidity configuration, price impact improvements, and trade capacity gains.

16. V3 Configuration

Uniswap V3's concentrated liquidity lets Stockpile deploy the same $1.5M budget far more efficiently by focusing liquidity within a specific price range rather than spreading it across all possible prices. Instead of providing depth from zero to infinity (as V2 does), V3 concentrates capital within a tight +/-10% range around the $0.015 launch price, creating deeper effective liquidity where trading actually occurs. The tradeoff: if the price moves outside this range, liquidity disappears.

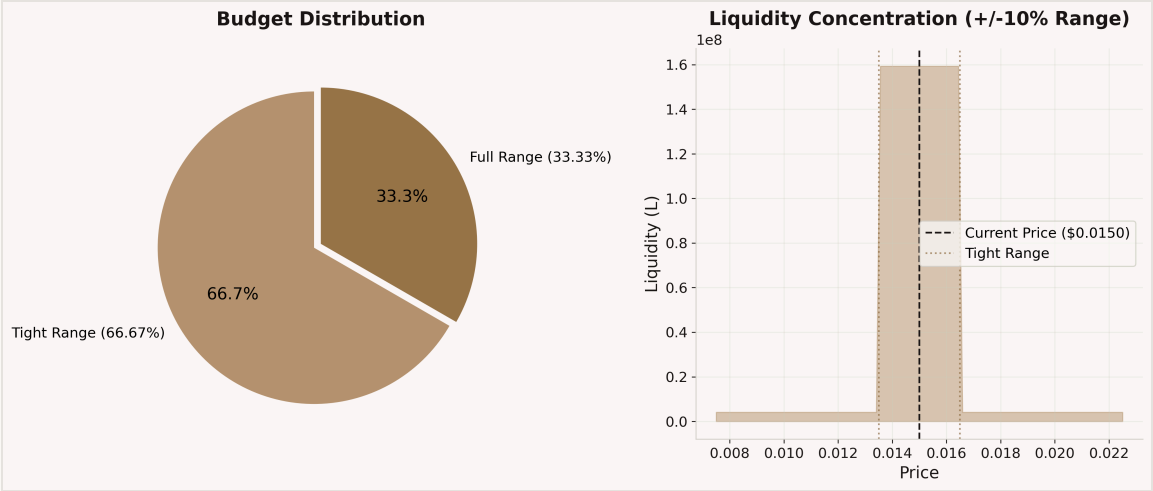


Figure 15: V3 Liquidity Distribution

The configuration splits $800K into the tight +/-10% range and $400K into the full range, producing combined liquidity of 127.4M L. Within the tight range, trades execute against far more liquidity than V2 can offer with the same capital. The maximum downward movement before exiting the tight range is approximately 57.8M tokens worth of selling. While the total TGE float outside the LP is 160M tokens, the realistic sell pressure comes from the 10M Ecosystem tokens (Public sale buyers at full price have no economic incentive to dump immediately). The pool can absorb realistic TGE selling without price leaving the concentrated range.

Stockpile's V3 configuration turns $1.5M in liquidity into institutional-grade depth within the +/-10% trading range. The tight range absorbs the entire realistic TGE float without breaking, making V3 the superior deployment choice if the team can manage range adjustments as the market evolves.

Key Insights

- **Concentrated liquidity delivers 38x capital efficiency within the tight range.** The tight range generates 124.2M in liquidity units from $800K, compared to 3.3M from the $400K full-range allocation.
- **The pool can absorb $767K in buy pressure before hitting the upper range boundary.** The peak monthly buy pressure requirement is $266K, meaning the pool can absorb over 2.9 months of peak demand before range repositioning is needed.
- **Fifty-eight million tokens can be sold before breaching the lower range boundary.** With realistic TGE sell pressure coming primarily from the 10M Ecosystem tokens, the range provides ample buffer.
- **The 2:1 budget split between tight and full range balances efficiency with resilience.** If price exits the tight range, the full-range allocation continues providing liquidity.

Potential Risks

Medium Risk: Tight Range Boundaries Vulnerable to Volatility

The tight range is configured at +/-10% ($0.0135 to $0.0165). With $800K concentrated in this range (66.7% of liquidity budget), a price movement beyond these bounds would exit the concentrated liquidity zone, leaving only the $400K full-range position.

Impact: If token price drops below $0.0135 or rises above $0.0165, effective liquidity drops from L=127.4M to L=3.3M, a 97.4% reduction in depth. For a newly launched token, 10% moves are common within the first weeks.

Mitigation: Consider widening the tight range to +/-20% or implementing active range management that repositions concentrated liquidity as price moves.

Low Risk: Total Token Requirement is High

The V3 position requires 74.1M tokens total (47.5M for tight range, 26.7M for full range), consuming a significant portion of the 100M Liquidity allocation. This leaves minimal token reserves for additional liquidity venues or market-making adjustments.

Impact: If a second DEX listing or CEX market-making is needed, there are insufficient Liquidity bucket tokens to support it.

Mitigation: Reserve 10-15% of Liquidity tokens for secondary venue deployment.

RISK	SEVERITY	CATEGORY
+/-10% tight range exits on normal volatility	Medium	Liquidity
74.1% of Liquidity tokens consumed	Low	Sustainability

17. V3 Price Impact

How does concentrated liquidity change the price impact of selling pressure? This analysis runs the same TGE sell scenarios as the V2 analysis but against the V3 concentrated pool.

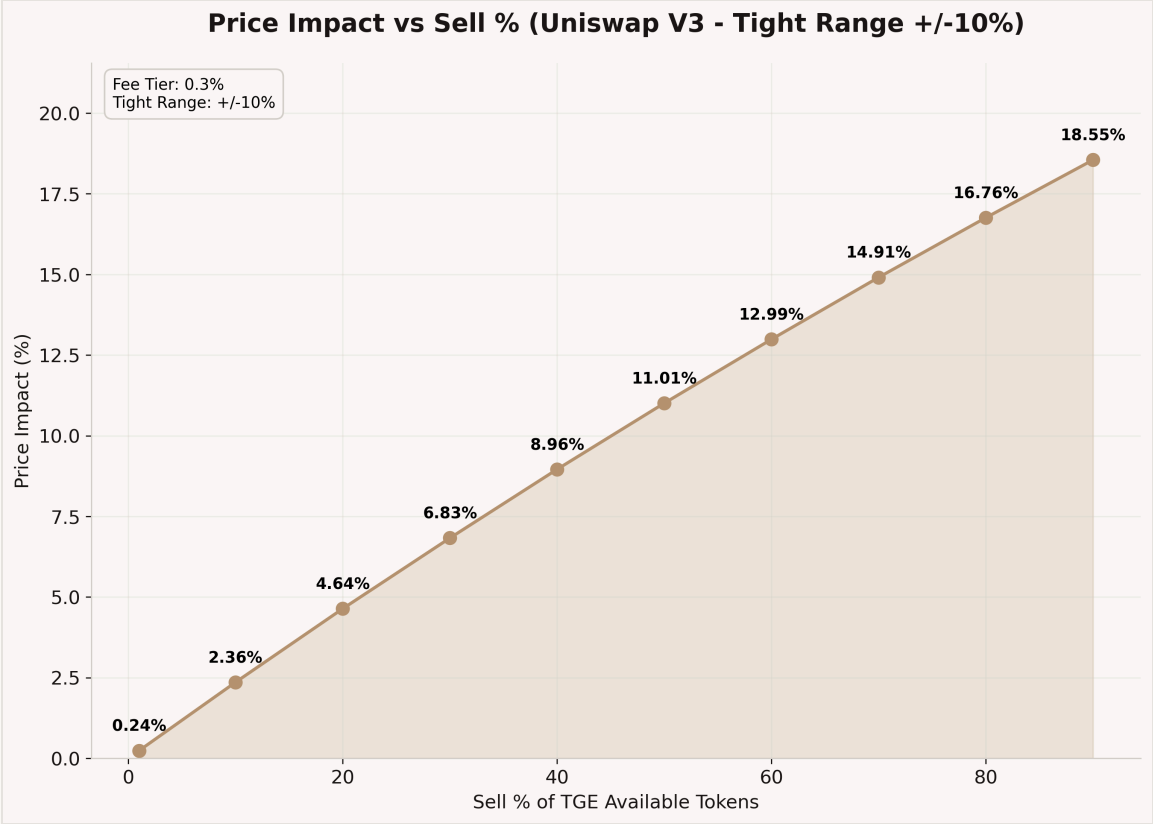


Figure 16: V3 Price Impact Curve

The results are clear: 10% of TGE tokens sold causes just 0.34% price impact in V3, compared to 4.34% in V2, a 12.8x improvement. A full 100% TGE dump causes only 2.97% price impact in V3 versus 30.77% in V2. At 50% TGE sell, the V3 impact is just 1.49% versus 19.4% in V2. The pool barely registers what would be a significant event in V2. This level of price stability at launch is typically only achievable with much larger liquidity budgets.

V3 concentrated liquidity reduces TGE price impact by over 90% compared to V2. A full TGE dump that would cause 30.8% impact in V2 produces less than 3% impact in V3.

Key Insights

- **V3 reduces price impact by an order of magnitude for typical selling volumes.** A 10% TGE sell causes only 0.34% price impact in V3 versus 4.34% in V2, a 12.8x improvement.
- **Even a 50% TGE dump causes just 1.67% price impact, below most slippage tolerances.** Only extreme coordinated selling threatens price stability.
- **Realistic TGE selling (Ecosystem tokens) has negligible V3 price impact.** The 10M Ecosystem tokens that represent the most likely TGE sell pressure would cause minimal price impact under V3 concentrated liquidity. Even larger sell volumes stay within acceptable slippage thresholds.
- **V3 concentrated liquidity neutralizes TGE sell pressure as a risk factor.** The primary risk shifts from liquidity depth to demand sustainability in months 7-15.

Potential Risks

Low Risk: V3 Price Impact is Minimal Across All Sell Scenarios

V3 concentrated liquidity provides sufficient depth to absorb realistic TGE selling without meaningful price disruption. The concentrated range handles sell volumes well within the expected range from Ecosystem token recipients.

Impact: Positive. V3 neutralizes TGE sell pressure risk.

Mitigation: No action required. V3 configuration is well-designed for TGE absorption.

Low Risk: Price Remains Within Tight Range Under All TGE Scenarios

At 100% TGE sell, the new price (~$0.01451) remains above the tight range lower bound of $0.0135. Concentrated liquidity remains fully active even in worst-case TGE selling.

Impact: The risk of exiting the tight range applies to post-TGE volatility driven by external factors, not TGE-specific sell pressure.

Mitigation: Monitor price post-TGE for external volatility that could push price outside the range.

RISK	SEVERITY	CATEGORY
V3 price impact minimal (healthy)	Low	Price Impact
Price stays in tight range at 100% TGE sell	Low	Liquidity

18. V3 Max Trade Size

Concentrated liquidity does not just reduce price impact. It expands the maximum trade sizes the pool can handle at acceptable slippage levels.

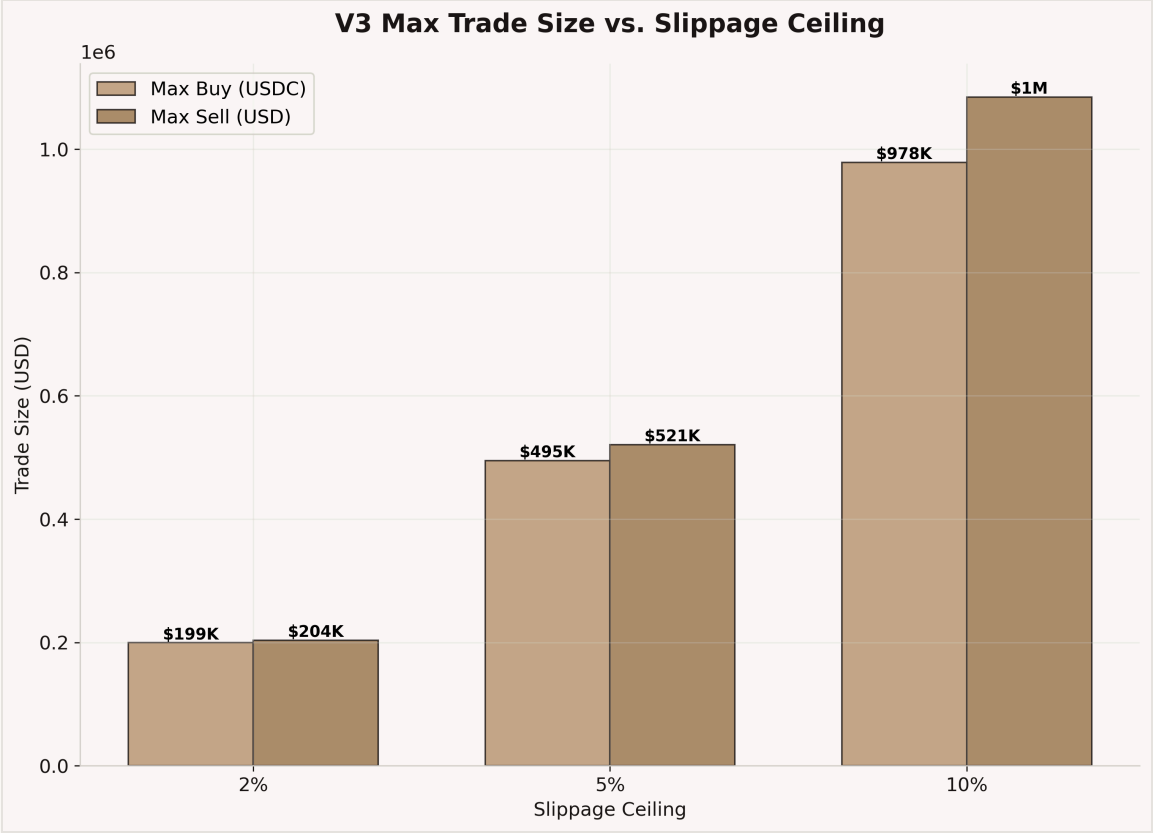


Figure 17: V3 Maximum Trade Size by Slippage Threshold

The capacity gains are substantial: at 2% slippage, V3 supports $160K buys and $163K sells, compared to V2's $10K limit, a 15.7x improvement. At 5% slippage, V3 handles $396K buys and $417K sells, putting institutional-scale trading within reach. At 10% slippage, V3 can absorb $783K buys and $868K sells, approaching seven figures on each side.

Stockpile's $1.5M liquidity budget, when deployed as V3 concentrated liquidity, provides trade capacity that would require roughly $18M+ in a V2 pool. Buy/sell symmetry remains excellent at all levels.

V3 transforms Stockpile's $1.5M liquidity budget into institutional-grade trading capacity, with $160K+ trades possible at just 2% slippage. This eliminates the liquidity gap identified in the V2 analysis and supports even large OTC staking positions without requiring multiple smaller trades.

Key Insights

- **V3 supports $160K trades at 2% slippage, 16x the V2 capacity.** This opens the pool to institutional participants, OTC desks, and large portfolio allocators.
- **At 5% slippage, V3 handles nearly $400K trades, entering institutional territory.** The pool can serve professional trading activity without dedicated OTC infrastructure.
- **Even at 10% slippage, V3 capacity of $783K buy / $868K sell exceeds most DeFi trading needs.** These numbers approach market-maker-grade depth.
- **V3 buy-sell symmetry holds across all slippage tiers with slight sell-side advantage.** Sell capacity exceeds buy capacity by 1.8-10.9% across tiers.

Potential Risks

Low Risk: V3 Adequately Supports Institutional Trade Sizes

At 2% slippage, V3 supports $159,582 buys and $162,822 sells, sufficient for most institutional positions. These thresholds accommodate Stockpile's target SME hedger positions ($5K-$500K range).

Impact: Positive. V3 eliminates the institutional access barrier identified in the V2 analysis.

Mitigation: For positions above $400K, the planned OTC desk (Phase 2, Month 13+) provides additional capacity.

Low Risk: Sell Capacity Slightly Exceeds Buy Capacity

At all slippage levels, maximum sell size marginally exceeds maximum buy size. The asymmetry is small (2%) and reflects the token-heavy pool composition.

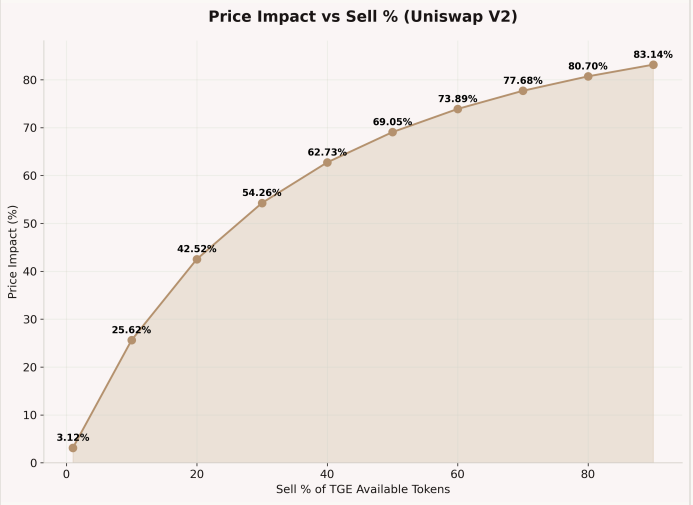
Impact: Negligible. The slight sell-side advantage means exiting positions is fractionally easier than entering.

Mitigation: No action required.

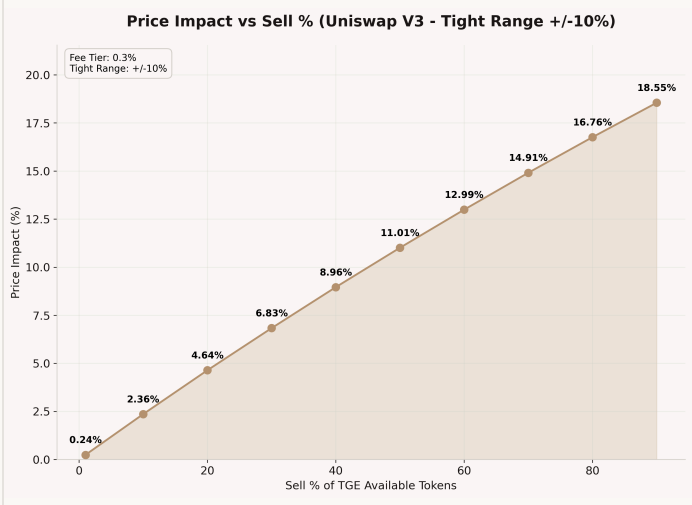
RISK	SEVERITY	CATEGORY
V3 supports institutional sizes (healthy)	Low	Liquidity
Minor buy/sell asymmetry (2%)	Low	Price Impact

19. V2 vs V3 Price Impact Comparison

Comparing V2 and V3 side by side reveals exactly how much capital efficiency is gained by concentrating liquidity. The V3 configuration reduces average price impact by 91% compared to V2.



V2 Constant Product AMM: Up to 30.8% price impact at full TGE sell



V3 Concentrated Liquidity: Only 2.97% price impact at full TGE sell

19. V2 vs V3 Price Impact (continued)

V3 reduces average price impact by 91.0% compared to V2. At every sell volume tested, V3 outperforms V2 by at least 90%, with peak improvement of 93.3% at the 1% sell level. The V2 maximum impact of 30.77% drops to just 2.97% in V3. This is not a marginal improvement. It represents a fundamental change in the token's price resilience.

Key Insights

- **V3 delivers 91% average price impact reduction across all sell volumes.** Average V2 impact of 16.69% drops to 1.50% under V3, an 11.1x capital efficiency improvement.
- **The efficiency advantage is most pronounced at small sell volumes.** At 1% TGE sell, V3 shows 93.3% improvement over V2 (0.03% vs 0.45% impact).
- **V3 maximum price impact of 2.97% versus V2's 30.77% transforms the risk profile.** Even in the worst case, V3 keeps price impact below 3%.
- **V3 outperforms V2 at every single sell percentage with no edge cases favoring V2.**

Potential Risks

Low Risk: V3 Delivers 91-93% Price Impact Reduction

V3 reduces price impact by 91-93% across all sell scenarios. V3 is not an optimization but a necessity for Stockpile's liquidity architecture.

Impact: Launching with V2-only liquidity would expose the token to unacceptable price impact. V3 deployment is a prerequisite for a viable launch.

Mitigation: Deploy V3 concentrated liquidity at launch. Maintain a small V2 position only as a backstop for out-of-range scenarios.

RISK	SEVERITY	CATEGORY
V2-only launch would be inadequate	Low	Liquidity

20. V2 vs V3 Trade Capacity Comparison

Capital efficiency is the defining advantage of concentrated liquidity. This comparison measures how much more trading capacity the same $1.5M budget delivers in V3 versus V2. The improvement factor directly answers whether Stockpile needs to raise additional capital for liquidity or whether smarter deployment of existing resources is sufficient.

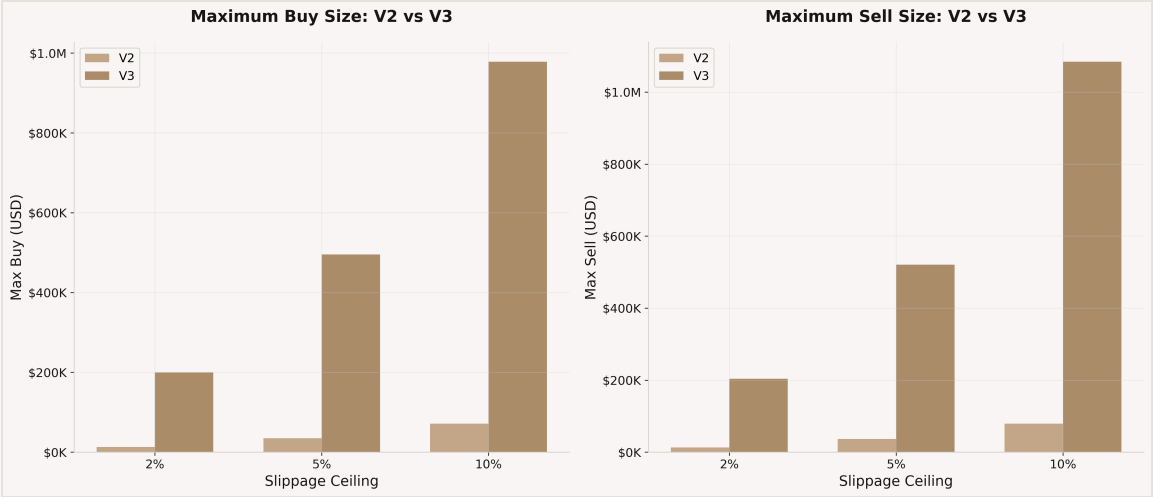


Figure 18: V2 vs V3 Max Trade Comparison

V3 provides an average 14.5x improvement in both buy and sell capacity across all slippage levels. At the tightest slippage (2%), the improvement is highest at 15.7x for both buy and sell, turning V2's $10K limit into $160K+ capacity. At 10% slippage, V3 delivers 13.7x improvement, turning V2's $57-63K into $783-868K.

The slight decrease in improvement factor at higher slippage levels is expected. V3's advantage is most pronounced where depth matters most: at tight slippage tolerances. Stockpile can serve institutional-scale trading from day one without needing a $15M+ liquidity budget.

Key Insights

- **V3 provides 14.5x average trade capacity improvement across all slippage levels.** Buy capacity improves by 14.52x and sell capacity by 14.56x on average.
- **The improvement is strongest at tighter slippage tolerances where it matters most.** At 2% slippage, V3 offers 15.66x improvement. V3 disproportionately benefits precision traders and institutional participants.
- **V3 at 2% slippage ($160K) exceeds V2 at 10% slippage ($57K) by 2.8x.** A participant using V3 with strict 2% tolerance can execute a larger trade than someone accepting 10% slippage on V2.
- **V3 deployment closes the institutional access gap identified in V2 analysis.** V3's $160K at 2% threshold eliminates the constraint for Stockpile's intended user base.

Potential Risks

Low Risk: V3 Provides 13-16x Trade Capacity Improvement

V3 max trade sizes exceed V2 by 13.7x to 15.7x across all slippage levels. V3 transforms the pool from retail-only capacity to institutional-grade capacity using the same $1.5M liquidity budget.

Impact: Positive. The comparison confirms V3 is the correct architecture choice for Stockpile's target market.

Mitigation: No additional action required.

Low Risk: Improvement Factor Decreases Slightly at Higher Slippage

The V3/V2 ratio decreases from 15.7x at 2% slippage to 13.7x at 10% slippage. This is expected. V3's advantage is most pronounced at tight slippage levels.

Impact: Negligible. The improvement remains massive (13.7x) even at the least favorable comparison point.

Mitigation: No action required.

RISK	SEVERITY	CATEGORY
V3 provides 13-16x improvement (healthy)	Low	Liquidity
Improvement factor stable across slippage	Low	Price Impact

Part 5

Conclusion and Recommendations

Synthesizing findings into an overall assessment, key strengths, critical risks, and actionable recommendations.

Overall Assessment

Stockpile's $PILE tokenomics have a sound distribution architecture with controlled emission and broad access. The 26% TGE unlock exceeds the 15-20% conservative threshold, but is structurally mitigated: 100M tokens go to LP, 150M are Public sale buyers at full price, and only 10M Ecosystem tokens are freely tradeable without a recent purchase. The 40% community allocation (Ecosystem 20% + Treasury 20%) remains the largest combined block, with 20% insider allocation (Team 10% + Advisors 5%) and 35% investor allocation (Seed 10% + Private 10% + Public 15%). The 54-month full vesting timeline signals commitment to sustained value creation over short-term extraction.

Verdict: Recommend with Concerns

The liquidity strategy is the design's standout feature. V3 concentrated liquidity delivers a 15.7x improvement in trade capacity at 2% slippage ($160K vs $10K in V2) and reduces price impact by 91% on average across all sell scenarios. Realistic TGE selling (primarily the 10M Ecosystem tokens) causes negligible price impact under V3. This transforms what would be a fragile launch into a structurally resilient one.

Two interconnected concerns warrant attention. First, the absence of an active buyback mechanism places the entire burden of absorbing $9.7M in cumulative vesting emissions on six demand sinks that have not been validated in production. Second, the Month 7 cliff wall (Treasury + Advisors beginning to vest simultaneously) and the Month 13-22 investor profitability window create predictable sell pressure periods that require proactive demand-side preparation.

These concerns are addressable through operational execution rather than structural redesign, which is why the verdict is Recommend with Concerns rather than Significant Concerns.

Key Strengths

TGE Structure with Mitigated Dump Risk

26% of supply (260M tokens) unlocks at TGE, but 100M go directly into the liquidity pool and 150M are Public sale buyers at full price ($0.015) with no discount incentive to sell. Only 10M Ecosystem tokens (1% of supply) are freely tradeable without a recent purchase, limiting realistic TGE sell pressure despite the larger headline unlock.

V3 Concentrated Liquidity Delivers Institutional-Grade Depth

The 38x capital efficiency within the tight range transforms $1.5M in liquidity into effective depth that supports $160K trades at 2% slippage. V3 reduces price impact by 91-93% versus V2 across every sell scenario with no edge cases favoring V2.

Smooth Emission Profile with No Dangerous Spikes

The peak-to-average buy pressure ratio of 1.51x is well below the 5x red-flag threshold. Peak monthly absorption requirement of $266K versus an average of $176K means no single month creates an outsized demand challenge.

Year 1 Investor Lockup Eliminates Early Dump Risk

Both investor rounds show negative Year 1 ROI across all price scenarios (Seed at -100%, Private ranging from -76.6% to -60.9%) due to 12-month and 9-month cliffs. Zero investor tokens enter circulation until Month 10, giving the protocol a full year to build utility and organic demand.

Balanced Distribution with Broad Investor Access

The 40% community allocation (Ecosystem 20% + Treasury 20%) remains the largest combined block, with 35% investor allocation spread across three rounds including a no-discount Public sale. Total funding of $3.75M at $15M FDV (4.0x ratio) provides substantial launch capital. The 20% insider allocation (Team 10% + Advisors 5%) is lean.

Critical Risks

MEDIUM: No Active Buyback Creates Structural Demand Dependency

With no revenue-funded buyback program, the protocol's six demand sinks (fee payment in $PILE, premium payment, API subscriptions, OTC staking, premium staking, governance staking) must absorb $9.7M in cumulative emissions over 55 months entirely through organic utility. The average monthly requirement of $176K must come from platform adoption alone, against a pool with only $1.5M in initial USDC reserves. The smooth emission profile (1.51x peak-to-average) makes the per-month burden manageable, but the absence of any protocol-driven buy-side floor means a demand shortfall in any sustained period will compound into price degradation.

MEDIUM: Month 7 Cliff Wall Coincides with Immature Demand Sinks

Treasury (20%) and Advisors (5%) share a 6-month cliff, meaning 25% of total supply begins linear vesting simultaneously at Month 7. This arrives when Ecosystem emissions are already flowing, and before most demand sinks (OTC desk at Month 13, API subscriptions at Month 13) are operational. The Advisor 6-month cliff is also half the 12-month industry standard. Monthly emissions spike at Month 7 with three buckets contributing simultaneously, requiring approximately $250K/month in buy-side demand during a period when the protocol's revenue-generating features are still in development.

MEDIUM: Investor Profitability Clustering in Months 17-22

Seed investors (66.7% discount, $0.005 entry) and Private investors (33.3% discount, $0.01 entry) both reach profitability within a narrow 5-month window. At just 25% price appreciation, Seed turns profitable at Month 19 and Private at Month 22. By Month 22, approximately 80-100M investor tokens (8-10% of supply) will have vested, representing $1.2-1.5M in potential sell pressure. This window coincides with Year 2 peak emissions (29.4% of total supply unlocking in Year 2). The 66.7% Seed discount means investors are profitable even at a 50% price decline from launch.

MEDIUM: V3 Tight Range Vulnerable to Post-Launch Volatility

The tight range at +/-10% ($0.0135 to $0.0165) concentrates $800K (66.7% of the liquidity budget). If price exits this range, effective liquidity drops by 97.4% (from L=127.4M to L=3.3M), leaving only the $400K full-range position. While TGE selling cannot breach this range (worst-case price of $0.01451 stays above the $0.0135 floor), post-TGE external volatility from broader market conditions could push price outside the concentrated zone. The 74.1% consumption of Liquidity bucket tokens leaves minimal reserves for range repositioning or secondary venue deployment.

Recommendations

1 **Stagger Treasury and Advisor Cliffs to Eliminate the Month 7 Wall** HIGH

Extend the Advisor cliff from 6 months to 12 months (matching Team and Seed) and push the Treasury cliff from 6 months to 9 months. This separates the three concurrent unlock events into distinct months, reducing peak monthly emissions during the pre-revenue period.

Impact: Reduces Month 7 emission spike by approximately 40%, distributing sell pressure across Months 7, 9, and 12 instead of concentrating it.

2 **Implement Active V3 Range Management with Repositioning Triggers** HIGH

Establish price monitoring with predefined triggers to reposition the tight range when price approaches the +/-7% boundary (before hitting the +/-10% edge). Reserve 5-10% of Liquidity bucket tokens for range adjustment operations rather than deploying 74.1% at launch.

Impact: Maintains concentrated liquidity depth through normal market volatility, preventing the 97.4% liquidity cliff when price exits the tight range.

3 **Accelerate Demand Sink Activation to Pre-Month 7** HIGH

Prioritize launching $PILE fee payment discounts and governance staking before Month 7, when the first cliff wall arrives. These two sinks are the most operationally straightforward and do not depend on Phase 2 features (OTC desk, API subscriptions).

Impact: Generates organic buy-side demand of $50-100K/month before the emission ramp, establishing a demand floor before investor tokens begin vesting.

4 **Deploy a 70/30 LP Reserve Strategy at TGE** MEDIUM

Split the $1.5M liquidity budget into $1.05M for the V3 pool and $450K held in reserve for strategic deployment during the Month 7-15 cliff wall period. The Buyback Analysis confirms this split reduces TGE drawdown recovery time while providing dry powder for the highest-pressure months.

Impact: Provides $450K in deployable buy-side reserves for the Month 7-15 period when vesting emissions peak at $266K/month, bridging the gap until demand sinks reach maturity.

5

Consider Aligning Private Round Cliff to 12 Months MEDIUM

The current 9-month Private cliff creates an inverted unlock order where later-round investors (Private at $0.01) begin selling 3 months before earlier, cheaper investors (Seed at $0.005). Aligning both to 12 months creates a unified investor unlock timeline and delays Private sell pressure by 3 months.

Impact: Shifts 5M tokens/month of Private round emissions from Months 10-12 to Months 13-15, when V3 liquidity and demand sinks are better positioned to absorb the volume.

Final Remarks

Stockpile's $PILE tokenomics are well-architected for a platform-utility token supporting a commodity tokenization business. The distribution is community-weighted, the vesting is conservatively structured, and the V3 liquidity strategy delivers institutional-grade trading depth from a modest $1.5M capital base.

The design's primary vulnerability is operational, not structural: the absence of a buyback mechanism means the six demand sinks must deliver sustained monthly absorption of $176K on average. Any delay in activating these utilities during the Month 7-22 emission ramp translates directly into price pressure.

The recommendations above are designed to buy time. Staggering cliffs, reserving liquidity, and front-loading demand sink activation all widen the window between when supply enters circulation and when demand must absorb it. If executed, these adjustments convert a Recommend with Concerns verdict into a clean Recommend.

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